

Nanopore for Educators eBook

**A community-developed guide to Nanopore sequencing in the
classroom**

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None

1. Nanopore for Educators eBook

A community-developed guide to Nanopore sequencing in the classroom

1.1 Welcome

This eBook is a community-authored guide to integrating [Oxford Nanopore](#) DNA sequencing into hands-on educational experiences. For around \$2000 in startup costs, students can start sequencing DNA using the same tools as cutting-edge researchers.

Nanopore Quick Start

Learn why Nanopore sequencing is the most classroom-friendly platform for research-grade DNA sequencing and course-based research experiences.

→ [Getting started](#)

Save time and money

Curated equipment and reagent list, cost estimates, and planning guides.

→ [Time and Cost Planning](#)

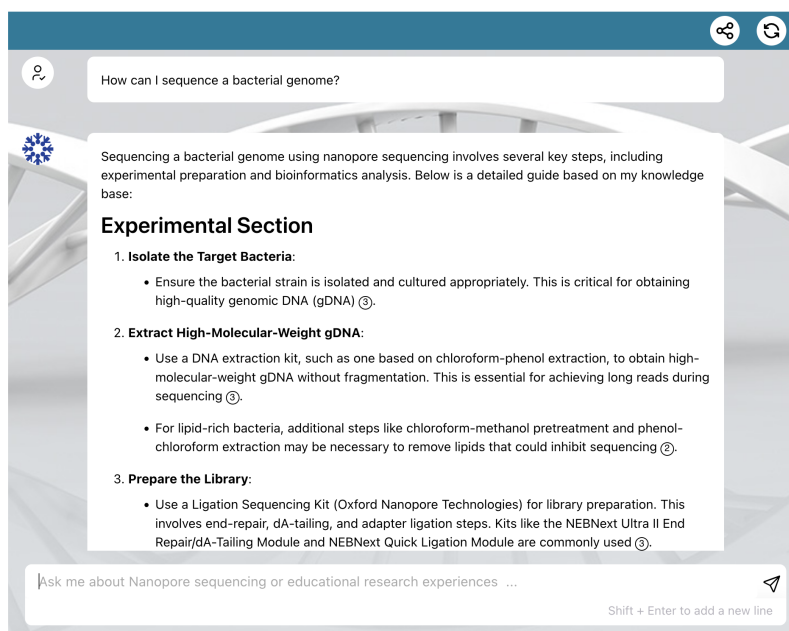
Highly-annotated protocols

Educator-friendly protocols combine tips and detailed explanations.

→ [16S Shoc-ome protocol](#)

Questions answered

Join our community! Get advice from an expert or an AI.



→ [Ask a question](#)

1.1.1 Why a guide for educators?

This guide is an educator-focused introduction to Nanopore DNA sequencing. Its primary goal is to share resources and approaches that make it easier to use Nanopore technology in the classroom, especially (but not exclusively) for course-based undergraduate research experiences (CUREs). Importantly, we aim to ensure that low-cost, easy-to-use DNA sequencing is accessible to institutions with fewer resources and to communities that haven't had full access to Science, Technology, Engineering, and Math (STEM) participation.

As a guide for educators, the content of this site is curated to focus on experiments and techniques well-suited for the classroom. We emphasize approaches that minimize costs and complexity, rather than those requiring the most well-equipped labs. Additionally, we aim to feature adaptable teaching resources (e.g., slides, lessons, protocols, videos, advice) to give educators a head start in developing their own teaching materials. The site is versioned to provide educators with a stable resource to rely on, even as updates are made.

1.1.2 What this guide is not

There are a few things to keep in mind when making use of this guide:

- This guide is not intended to replace or supersede official protocols or warranty agreements.
- It is not a publication of [Oxford Nanopore Technologies](#). All content is contributed and curated by community members. While protocols and recommendations are tested, users assume all risks. We include, link to, and cite resources, including peer-reviewed articles for further exploration.
- The guide is not a comprehensive exploration of all the capabilities of the Nanopore sequencing. While we include experiments and techniques that can be successfully used by a wide range of educators, we do not cover every potential application.

1.1.3 Nanopore Network

We are a community you can join!

Connect with other like-minded educators interested in using Nanopore in the classroom. Join our Slack channel for realtime discussion and connect with us through [QUBES](#), where we organize semester-long "faculty mentoring networks." These are regular meetings of faculty members who share knowledge and experience about integrating Nanopore into teaching, with a focus on CUREs.

[Join the Slack](#)
[Join on QUBES](#)

1.1.4 Ask a question

Our experimental AI chatbot is trained on a variety of Nanopore reference materials, and peer-reviewed publications on Nanopore sequencing and undergraduate education. Ask a question to receive a synthesized answer, complete with references to the original sources. Please verify your understanding with a trusted source, as no AI bot can guarantee 100% accuracy.

1.1.5 Citation, licensing, and reuse

- **Citation:** Coming soon
- **Licensing:** Unless otherwise noted, all information developed for and distributed through this site not separately attributed or under the terms of another license or usage agreement (e.g. publications and contributed resources) may be considered licensed under [Creative Commons BY 4.0 Attribution 4.0 International](#). All information is presented for educational use and without warranty.

1.1.6 Funding acknowledgment, and declared interests

This material is based upon work supported by the National Science Foundation under Grant No. (DUE:2216349) as a Level-1 multi-institutional IUSE (Institutional and Community Transformation) project, we have assembled a diverse team of investigators to develop and test infrastructure to support genome

science at three MSIs: New York City College of Technology, Spelman College, the University of Puerto Rico-Río Piedras. It is led by the Cold Spring Harbor Laboratory [DNA Learning Center](#).

This site exclusively highlights the potential of Oxford Nanopore Sequencing because currently, it is the only DNA sequencing platform we believe provides an accessible and affordable hands-on platform for DNA sequencing by pre-college and college students. The authors declare no financial conflict of interest with Oxford Nanopore. At times, Oxford Nanopore has provided the authors and other educators with free or discounted materials exclusively for use in training and education as part of its educator discount program.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used OpenAI ChatGPT to generate summary content for various sections and for grammatical correction, proofreading, and formatting. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.



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2. Get Started

2.1 Getting Started with Nanopore Sequencing

2.1.1 Why Nanopore sequencing is promising for education

For decades, education leaders have emphasized the need to expand and strengthen the STEM workforce. Among the many strategies to achieve this goal, providing students with authentic research experiences has proven to be the most effective. Research participation significantly increases the likelihood that students will persist in and complete STEM degrees. Yet, one major challenge remains: ensuring that all STEM students, regardless of their institution or background, have access to meaningful research opportunities.

Nanopore DNA sequencing is uniquely positioned to address this challenge. This portable, low-cost technology has democratized access to DNA sequencing, placing powerful research tools within reach of pre-college and college classrooms. The same technology used by top researchers can now empower students to investigate their own scientific questions, fostering deeper engagement and understanding.

Integrating DNA sequencing into STEM curricula unlocks numerous benefits:

- **Active Learning:** Students gain hands-on experience with cutting-edge technology, transforming abstract concepts into tangible discoveries.
- **Skill Development:** From molecular biology to data science, students build multidisciplinary skills critical for the modern STEM workforce.
- **Ethical Research Practices:** Teaching responsible research conduct equips students to navigate complex challenges in the scientific and societal spheres.

National networks of bioscience educators and industry professionals have demonstrated the transformative potential of course-based research experiences. Scalable and engaging, DNA sequencing-based research experiences can reach large numbers of students, providing an equitable pathway into science. With coordinated efforts, we can bring these opportunities to classrooms across the country, ensuring that all students have a chance to engage meaningfully with science.

Accessible DNA sequencing is more than just a tool for examining the blueprint of life—it's a roadmap for advancing STEM education and cultivating the next generation of scientists.

2.1.2 What is Oxford Nanopore sequencing?

Oxford Nanopore sequencing is an innovative technology that allows scientists and students to sequence DNA by passing molecules through tiny nanopores embedded in a membrane. As the nucleic acids move through the nanopores, they create electrical disruptions, which are analyzed in real-time to determine the sequence. This method enables quick, portable, and cost-effective sequencing—ideal for educational settings.

For a beginner-friendly overview, watch the short introductory video [How nanopore sequencing works](#)  on the Oxford Nanopore Technologies YouTube channel. It explains the basic principles of the technology.

How Nanopore sequencing works

Key features of Oxford Nanopore sequencing include:

- **Low Cost:** Oxford Nanopore devices have significantly lower startup costs compared to traditional platforms, making advanced DNA sequencing more accessible.
- **Student-friendly:** With a little training, a high school or undergraduate can do every step of most sequencing workflows. With some experience, the entire Nanopore sequencing workflow is something students can reliably operate.
- **Portability:** Devices like the MinION are small enough to fit in your pocket, making them accessible for classroom use. Sequencing can be done in the classroom, rather sent off to some remote institution. This builds local capacity and turns institutions that may not have previously participated into potential genome sequencing centers.
- **Real-Time Data:** With real-time data streaming, educators can engage students in experiments where results are visible immediately, enhancing the learning experience.
- **Long Reads:** The technology excels at sequencing long DNA fragments, providing insights into complex genomic regions that are difficult to analyze with traditional methods.

Oxford Nanopore sequencing has been used in groundbreaking projects, from sequencing genomes in space to tracking viral outbreaks like SARS-CoV-2. Its versatility and affordability make it a powerful tool for transforming biology education, enabling students to engage with modern research techniques and develop essential skills in molecular biology and data science.

How accurate is Nanopore sequencing?

>99% accuracy and improving

In the early development of the technology more than a decade ago, Nanopore sequencing had significant issues with sequence quality. That is no longer the case. The quality of Nanopore data is not only suited for the classroom, but in real research application — for example in [the completion of the human genome](#). In addition to the chemistry improvements that have helped, [error correction methods](#) have further improved the quality of the data you can generate.

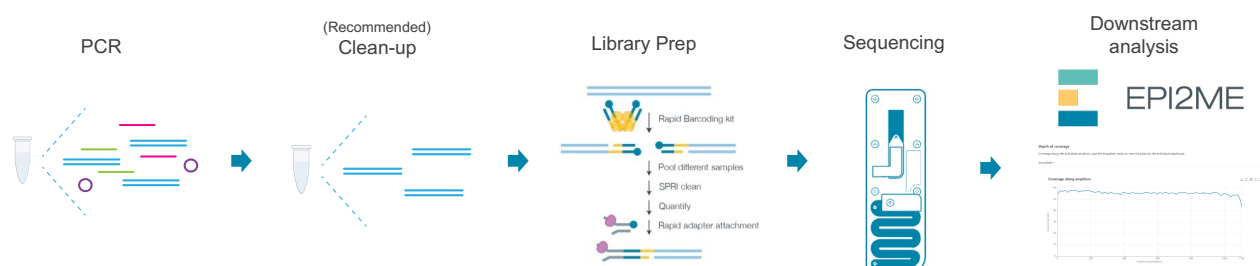
Oxford Nanopore [makes these claims](#) about the accuracy of its platform. We also recommend you examine the latest peer-reviewed research. Here are some helpful papers:

- Stevens, B.M., Creed, T.B., Reardon, C.L. et al. **Comparison of Oxford Nanopore Technologies and Illumina MiSeq sequencing with mock communities and agricultural soil.** Sci Rep 13, 9323 (2023). <https://doi.org/10.1038/s41598-023-36101-8>
- Cuber, P., Chooneea, D., Geeves, C., Salatino, S., Creedy, T. J., Griffin, C., Sivess, L., Barnes, I., Price, B., Misra, R. (2023). **Comparing the accuracy and efficiency of third generation sequencing technologies, Oxford Nanopore Technologies, and Pacific Biosciences, for DNA barcode sequencing applications.** Ecological Genetics and Genomics, 28, 100181. ISSN 2405-9854. <https://doi.org/10.1016/j.egg.2023.100181>.
- Zhang, T., Li, H., Jiang, M., Hou, H., Gao, Y., Li, Y., Wang, F., Wang, J., Peng, K., Liu, Y.-X. (2024). **Nanopore sequencing: flourishing in its teenage years.** Journal of Genetics and Genomics, 51(12), 1361–1374. ISSN 1673-8527. <https://doi.org/10.1016/j.jgg.2024.09.007>.

What does a Nanopore sequencing workflow look like?

Example DNA Barcoding Workflow

We routinely do complete experiments—sample collection to data analysis—in one day. Several factors impact how much time is needed, but this example is fairly representative:



PCR amplicon sequence (as presented on Oxford Nanopore protocol page).

Basic Steps

- **PCR:** 1-3 hours (there are PCR-free methods such as genome sequencing)
- **PCR clean up** (and, we'd recommend quantification): 60 min
- **Library prep** (adapters added and sample prepared): 45-60 min
- **Sequencing:** Useful data in 30-60 min, longer runs for additional data
- **Data analysis:** Realtime analysis can generate results in minutes, and many datasets (e.g. DNA barcoding) can be analyzed in under 20 minutes. Many analyses don't require any coding or complex bioinformatics skills.

2.1.3 Am I ready to bring Nanopore sequencing into my classroom?

To determine your readiness to integrate Nanopore sequencing into your classroom, consider the following self-assessment. Rate yourself for each consideration on a scale of 1–3:

- **1:** I'm just starting out or have limited experience/resources in this area.
- **2:** I have some experience/resources but may need additional preparation or support.
- **3:** I'm confident, well-equipped, or fully prepared in this area.

Keeping score

Use the checkboxes to keep track of your self-assessment scores.

1. Technical readiness

- **Access to Equipment:** Do you have the necessary devices (e.g., MinION sequencer, Flongles), molecular biology reagents (e.g., DNA extraction equipment, micropipettes, PCR machine), a laptop, and other essential peripherals?

1 - I don't yet have access to these tools.

2 - I have some access but may need additional equipment.

3 - I have full access to all required tools.

- **Laboratory Facilities:** Is your lab space equipped for basic molecular biology work (e.g., pipettes, centrifuges, PCR)?

1 - My lab is not set up for molecular biology or I don't have lab space.

2 - My lab has some resources but might need upgrades.

3 - My lab is fully equipped and ready.

2. Training and expertise

- **Familiarity with DNA Sequencing:** Do you have foundational knowledge of DNA sequencing techniques?

1 - I'm new to DNA sequencing.

2 - I have some understanding but may need additional training.

3 - I'm confident in my understanding of DNA sequencing.

- **Bioinformatics Knowledge:** Do you or your students have basic skills for analyzing sequencing data?

1 - I have no bioinformatics experience.

2 - I have some familiarity but need further support.

3 - I have sufficient knowledge to analyze sequencing data.

3. Budget and resources

- **Cost of Reagents:** Do you have funding for consumables like sequencing kits and flow cells?

1 - I don't have a budget for reagents yet.

2 - I have some funding but may need to secure more.

3 - I have a reliable budget for necessary materials.

4. Curriculum alignment

- **Course Goals:** Does integrating Nanopore align with your course’s learning objectives?

1 - I’m unsure how this fits into my course.

2 - I see potential alignment but need to refine objectives.

3 - Nanopore integration fits perfectly with my objectives.

- **Time Commitment:** Can your curriculum accommodate the time for sequencing experiments and data analysis?

1 - My schedule is too tight for this addition.

2 - I can make room but may need careful planning.

3 - My schedule can fully support this integration.

5. Student readiness

- **Skill Level:** Are your students prepared to engage with molecular biology and data analysis concepts?

1 - Students have minimal exposure to these concepts.

2 - Students have some familiarity but need more preparation.

3 - Students are well-prepared for these activities.

- **Engagement:** Are your students likely to find the experiments and research questions motivating?

1 - Students may find this unfamiliar or challenging.

2 - Some students might be engaged; others need motivation.

3 - Students will be highly motivated by these activities.

6. Community and support

- **Mentorship and Networks:** Are you part of an educator community that can support you in implementing Nanopore sequencing?

1 - I don’t have a network to rely on.

2 - I have some connections but would benefit from more support.

3 - I’m part of a strong community of educators.

7. Ethical and inclusive practices

- **Responsible Conduct of Research:** Are you ready to teach ethical considerations like data handling and equity in research?

1 - I’m unfamiliar with these practices.

2 - I have some knowledge but need more preparation.

3 - I’m confident in teaching these topics.

- **Inclusivity:** Can you design lessons that connect sequencing to student- and community-relevant topics?

1 - I’m unsure how to make lessons relevant.

2 - I have ideas but need more resources.

3 - I can confidently create relevant and engaging lessons.

What do my scores mean?

If you are mostly scoring **2 or above**, you are likely ready to bring Nanopore sequencing into your classroom. Even if you rated yourself lower in some areas, don’t worry—this guide is designed to provide the support, resources, and community you need to get started and build confidence.

Remember, teaching with new technology is a learning process for both you and your students. Embrace the opportunity to grow together while exploring the exciting world of Nanopore sequencing!

Training opportunities

Training Opportunities

In addition to the resources on this site, please bookmark the faculty training page of the [DNA Learning Center](#). We will post regular free training opportunities.

2.1.4 From absolute beginner to competent and confident

Embarking on the journey to integrate Nanopore sequencing into your classroom can feel daunting at first, but with the resources provided on this site, you'll be fully supported as you progress. Whether you're new to DNA sequencing or just unfamiliar with Nanopore technology, this site is designed to guide you step-by-step, helping you grow from your first experiment to confidently leading students through research experiences.

What this site offers

- **Getting started checklists**

These checklists will help you ensure you have the minimum equipment needed to begin, understand the time and costs involved, and plan effectively for your first experiments.

- **Demonstration experiments**

Explore classroom-friendly experiments with detailed protocols, tips, and troubleshooting advice. These experiments are great starting points, enabling you and your students to gain hands-on experience with Nanopore sequencing.

- **Bioinformatics guides**

Even if you have limited computational skills, our bioinformatics sections provide straightforward protocols to help you get set up. Step-by-step guidance ensures you can analyze sequencing data and interpret results with confidence.

- **Example datasets**

Use curated datasets to familiarize yourself with the outcomes of a typical experiment. These examples allow you to practice data analysis and understand what to expect when working with real sequencing results.

- **Teaching resources**

Discover examples of effective teaching practices and approaches. From lesson plans to strategies for engaging students, these resources will inspire you to integrate Nanopore sequencing into your curriculum in meaningful ways.

- **Community and peer support**

Connect with other educators who are also implementing Nanopore sequencing in their classrooms. Join virtual communities like the [QUBES Nanopore Network](#) and our [Slack group](#) to exchange ideas, share experiences, and troubleshoot challenges. Additionally, periodic opportunities for in-person training will allow you to build hands-on skills and grow your network.

A journey of growth

Remember, this is a journey—not a sprint. You don't need to master everything at once. Start small, with a single experiment or activity, and build from there. Over time, you'll develop skills in experimental techniques, data analysis, and instructional design.

As you progress, you'll gain confidence and competence, and your students will benefit from engaging in authentic research experiences. Whether you're teaching molecular biology, genetics, or data science, Nanopore sequencing can become a transformative tool in your classroom. Go at your own pace, and know that this site and its vibrant community are here to support you every step of the way.

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2.2 Ten Steps to Getting Started with Nanopore

Essential steps for getting started with Nanopore sequencing

1. Understand the Basics of Nanopore Sequencing

- Familiarize yourself with how Nanopore sequencing works, its capabilities, and its limitations.
- Review this book chapter: [The Current State of Nanopore Sequencing 2023](#)
- View Nanopore's video: [How to get started with nanopore sequencing and plan your experiment](#) 
- Review advanced research [Applications](#) making use of Nanopore sequencing.

2. Assess Your Curriculum Goals

- Define the objectives for integrating Nanopore sequencing into your classroom.
- Determine whether you will focus on specific applications (e.g., microbial diversity, genome assembly, or barcoding) or introduce students to the general principles of sequencing.
- Know benefits and strategies for introducing course-based research experiences.
- Find resources for developing undergraduate course-based research experiences: [Fear of the CURE: A Beginner's Guide to Overcoming Barriers in Creating a Course-Based Undergraduate Research Experience](#)

3. Plan and Budget for Equipment and Materials

- Use our [list of essential equipment](#) (e.g., [MinION](#) sequencer, magnetic racks, pipettes) and reagents (e.g., DNA extraction kits, sequencing kits).
- We recommend starting with the [16S demonstration experiment](#) or [DNA barcoding demonstration experiment](#) to get started. Each experiment details all the needed materials.
- Use educational discounts where available to minimize costs.
- Check with colleagues at your institution or nearby who may be using Nanopore and can provide you with support.

4. Set Up Online Accounts and Software

- Register for an [Oxford Nanopore Technologies account](#).
- Orders for Nanopore devices, [library kits](#), reagents and accessories at the [Nanopore store](#).
- You will likely need to coordinate with your purchasing department to setup shipping and payment.
- Orders can also be placed through an authorized vendor such as [Avantor](#); check with Nanopore on additional options.
- You will be able to purchase a [starter pack](#) to get your device.
- Explore free platforms like [Galaxy](#), [DNA Subway](#), and [CyVerse](#) for bioinformatics workflows.

5. Acquire Hands-On Experience

- Conduct trial runs of Nanopore workflows before teaching.
- Practice key steps such as DNA extraction, [library preparation](#), and sequencing.
- Use mock samples or pre-prepared datasets to familiarize yourself with bioinformatics tools.

6. Develop or Adapt Teaching Materials

- Prepare slides, handouts, and protocols tailored to your specific learning objectives.
- Include clear instructions for students on safety, experimental techniques, and data analysis.
- Incorporate assessments to evaluate students' understanding and progress.

7. Engage in Protocol Optimization

- Test protocols with your lab aides or instructional staff to identify potential bottlenecks.
- Adjust protocols for compatibility with your class size, time constraints, and available resources.

8. Plan Your First Experiment

- Choose a simple and achievable experiment (e.g., sequencing microbial DNA or barcoding plant species).
- Use small-scale experiments to build student confidence and refine your approach for larger projects.
- Layout how sequencing work will be divided between individual students, student groups, and instruction staff.

9. Leverage Community Resources

- Join educator-focused communities like [Join our QUBES site and network](#) and [Nanopore Network Slack](#).
- Attend faculty mentoring networks, webinars, or in-person training sessions to exchange knowledge and strategies with peers.

10. Evaluate and Iterate

- After your first experiments, evaluate student outcomes and overall workflow.
- Gather feedback from students and instructional staff to refine future experiments and lesson plans.
- Document your experience to build a robust and repeatable curriculum.

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2.3 Estimating time and costs to get started with Nanopore

Integrating Nanopore sequencing into a classroom or teaching laboratory offers many benefits. However, estimating the time and total costs required to get started is essential for successful planning and implementation. This section provides some tips and observations to help educators understand the resources and time needed to bring Nanopore sequencing to their students.

When estimating costs, educators should consider both initial and recurring expenses. These include purchasing essential equipment, consumables, reagents, and sequencing kits, as well as ensuring access to appropriate computational resources. Time estimates should account for equipment setup, instructor preparation, and the time required to guide students through laboratory protocols and bioinformatics analyses.

It's important to note that these are general guidelines, and several factors can influence both the time and cost of implementation. For example:

- Class size and the number of sequencing projects planned may affect the quantity of consumables and reagents required.
- Availability of existing laboratory equipment and computational resources can significantly reduce start-up costs.
- Instructor experience with molecular biology techniques and bioinformatics may influence preparation time and the level of support needed.
- Institutional purchasing agreements or discounts may reduce the costs of equipment and reagents.

2.3.1 Example prep and teaching activities and timings

Activities	Estimated Duration	Items/Activities to Add Up Costs(\$)	Notes
Background research on Nanopore sequencing	(1–2) days	Training workshops (free-\$ for travel or tuition)	Time for familiarization; consider workshops or online resources
Curriculum planning	(1–2) weeks	Lesson plans, assessments, teaching materials	Aligns activities with course objectives; includes material development
Protocol testing	(1) month	Reagents (\$\$\$), consumables (\$), staff time	Trial runs with aides or TAs; optimize protocols for classroom
Sample collection	(1) 2-hour lab section	Collection supplies (\$), travel (\$), and equipment (\$\$)	Depends on sample type (e.g., environmental, lab specimens)
DNA extraction	(1–2) 2-hour lab sections	Reagents (\$\$), consumables (\$), equipment (\$\$\$)	Includes DNA isolation and quality checks; depends on sample type
Library preparation	(1-2) 2-hour lab section	Sequencing kits (\$\$), consumables (\$), reagents (\$\$)	Laboratory staff or instructor pools samples and generates libraries
DNA sequencing	(1) 2-hour lab section	Flow cells (\$-\$), sequencing kits (\$\$), compute resources	Some samples may require 1-hour runs; others may need 24+ hours
Data analysis	(2) 2-hour lab sections	Software subscriptions (\$), cloud resources (free-\$)	Includes bioinformatics tools and interpretation; longer steps like <u>basecalling</u> occur outside class time
Evaluation and feedback	(1) week	Instructor time, survey tools, debriefing materials	Grading assessments, collecting feedback, and debriefing instructional staff

2.3.2 Example costs - Nanopore equipment

This table outlines the estimated minimal startup costs for purchasing your first set of Nanopore sequencing materials, assuming that you already have most basic molecular biology reagents and equipment. The potential education discounts are included for illustrative purposes only—contact Oxford Nanopore to inquire on available education discounts.

Item	Section/Class Quantity	Base Cost	Total at Base Cost	Education Discount Price*	Total w/Discount
MinION Startup Pack (MinION flow cell Flongle flow cells, sequencing kit)	1	\$2,000	\$2,000	\$1,600	\$1,600
Flongle and adapter	1	\$1,460	\$1,460	0 (included in startup pack)	
Magnetic rack	6	\$59	\$354	\$59	\$354
MacBook computer	1	\$2,800	\$2,800	\$2,400	\$2,400
Total Startup Costs			\$6,614		\$4,354

*The provided educational discounts are illustrative and may vary depending on institutional agreements or vendor promotions

2.3.3 Example costs - Nanopore reagents and consumables

This table provides an overview of the estimated costs for different Nanopore sequencing kit types and associated reagents when used in classroom settings. It includes base costs, costs per sample or reaction, and potential educational discounts. The table highlights the potential costs for a class of 24 students, assuming standard consumables and reagents.

16S Kit (6 x 24 samples) - Microbial diversity

Nanopore Sequencing Kit Type	Base Cost	Total per Sample/ Run	Education Discount Price*	Total per Sample/Run w/ Discount
Nanopore 16S Kit	\$900.00	\$6.25	\$400.00	\$2.78
Norgen DNA extraction kit and swabs (50 preps)	\$638.00	\$12.76	\$638.00	\$12.76
NEB LongAmp Taq (100 rxn)	\$161.00	\$1.61	\$128.00	\$1.28
Flongle (12 pack - 1 Flongle per run)	\$67.00	\$2.79	\$67.00	\$2.79
Consumables (\$4 per student)	\$96.00	\$4.00	\$96.00	\$4.00
Potential cost per student/run		\$27.41		\$23.61
Total for class of 24		\$657.88		\$566.63

Rapid Barcoding Kit (6 x 24 samples) - DNA barcoding and small genomes

Nanopore Sequencing Kit Type	Base Cost	Total per Sample/ Run	Education Discount Price*	Total per Sample/Run w/ Discount
Nanopore Rapid Barcoding Kit	\$699.00	\$4.85	\$110.00	\$0.76
Low-cost "homemade" DNA extraction (24 preps)	\$24.00	\$1.00	\$24.00	\$1.00
Low-cost Taq and PCR reagents (24 preps)	\$50.00	\$2.08	\$50.00	\$2.08
Flongle (12 pack - 1 Flongle per run)	\$67.00	\$2.79	\$67.00	\$2.79
Consumables (\$4 per student)	\$96.00	\$4.00	\$96.00	\$4.00
Potential cost per student/run		\$14.73		\$10.64
Total for class of 24		\$353.50		\$255.33

Ligation Kit (6 samples) - Small genomes

Nanopore Sequencing Kit Type	Base Cost	Total per Sample/ Run	Education Discount Price*	Total per Sample/Run w/Discount
Nanopore Ligation Kit	\$699.00	\$233.00	\$110.00	\$36.67
DNA extraction kit (example, NEB Monarch) 50 samples; 6 runs	\$517.00	\$20.68	\$413.60	\$16.54
NEB companion module (24 reactions); 6 runs	\$1,025.00	\$85.42	\$820.00	\$68.33
MinION flow cell (1 per run - assuming 2 runs, washed and reused)	\$750.00	\$375.00	\$450.00	\$225.00
Consumables (\$4 per student)	\$96.00	\$4.00	\$96.00	\$4.00
Potential cost per student/run (2 runs or non-barcoded samples, 24 students)		\$718.10		\$350.54
Total for class of 24		\$718.10		\$350.54

*The provided educational discounts are illustrative and may vary depending on institutional agreements or vendor promotions

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2.4 Start-up Checklists

To run a successful Nanopore sequencing experiment in your classroom, you'll need a combination of basic molecular biology tools, a Nanopore sequencing device, and computational resources. Below is an overview of the materials required, with notes on equipment we generally consider essential. Some specific recommendations also include links to products we have had good success with.



Warning

Check your protocol

These lists are generalizations. Review the specific protocols for your experiments to verify you have the needed supplies. Also note, that PPE recommendations are also specific to the materials and reagents you will use, as well as your institution's laboratory safety guidelines. Please follow those specific guidelines.

2.4.1 Equipment, reagents, supplies, and computer technologies lists

Personal Protective Equipment

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
PPE	Disposable gloves	Yes	In addition to reagent contact, recommended to avoid cross contamination	1 pair per student, lab aide, and instructor per session	Nitrile gloves	\$10–\$30 (100)
PPE	Lab coats		Ensures personal and sample safety	1 per student, lab aide, and instructor	Standard lab coats	\$20–\$50
PPE	Safety goggles		May be called for during sample collection or DNA extraction.	1 per student, lab aide, and instructor	Chemical splash goggles	\$5–\$20
PPE	Waste disposal bags		For safe disposal of biological waste	Varies	Biohazard disposal bags, especially needed if working with cultured microbial samples	\$10–\$30

Molecular Biology Equipment

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Equipment	Micropipettes (1-10 µl, 10-100 µl, 100-1000 µl)	Yes	For precise liquid handling in Nanopore protocols.	1 set per group + 1 for instructor	Adjustable-volume pipettes from reliable brands like Eppendorf or Gilson.	\$50–\$300
Equipment	Thermocycler	Likely	Supports 24+ barcoded samples. Gradient PCR capability is recommended for optimization.	1 per lab	Bio-Rad or Thermo Fisher systems are suitable for classroom use. Smaller units by miniPCR are inexpensive.	\$700–\$1,500
Equipment	Microcentrifuge	Yes	For 12-24 1.5 ml tubes. Smaller format for PCR tubes is helpful but not essential.	1 per lab	Tabletop models from Eppendorf or Thermo Fisher.	\$200–\$500
Equipment	Water bath or heat block	Yes	For incubation steps. Hotplates may be used for preparing electrophoresis gels.	1 per lab (some protocols benefit from additional units)	Heat blocks with adjustable temperature settings are versatile for multiple protocols.	\$150–\$500
Equipment	Microwave		Hotplates may also be used for preparing electrophoresis gels.	1 per lab	Standard lab microwave or hotplate.	\$50–\$1,000
Equipment	Electrophoresis tank and power source	Likely	Recommended for checking PCR products and, in some cases, DNA quantity and quality.	1 set per group or 1 or more communal sets	Compact electrophoresis systems are recommended. See sets such as Carolina and MiniPCR QP-1600-01 .	\$200–\$500
Equipment	Electrophoresis documentation setup	Likely	UV light source for gel visualization; camera setup (can include a cell phone with proper filters).	1 per lab	Blue light transilluminators are a safe alternative to UV setups.	\$300–\$500
Equipment	Glassware for agarose gels (100 ml cylinder, 200 ml Erlenmeyer flask)	Likely	Required for gel preparation.	1 set per lab	Durable glassware from lab supply companies.	\$20–\$50
Equipment	Magnetic rack	Yes	For magnetic bead cleanups. Should support 1.5–2.0 ml tubes.	1 per group and 1 or more for lab aid and instructor	Sergilabs offers affordable options. Amazon link	\$30–\$100

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Equipment	Ice bucket(s)/ styrofoam containers	Yes	For keeping reagents and samples cold during preparation.	1 per group and several for lab use	Reusable or disposable options depending on budget.	\$10–\$50
Equipment	-20°C non-defrosting freezer	Yes	Essential for storing reagents and DNA samples. Low-cost domestic units can be substituted (\$200-300) must be non-defrosting .	1 per lab	Lab-grade non-defrosting freezers are essential to prevent degradation of sensitive materials.	\$1,000–\$2,000
Equipment	4°C fridge	Yes	For short-term reagent and sample storage as well as storage of <u>flow</u> cells. Low-cost domestic units can be substituted (\$200-300).	1 per lab	Lab-grade refrigerators preferred for consistency.	\$1,000–\$2,000
Equipment	Qubit fluorometer		For DNA quantification. Many protocols will work with estimates provided by electrophoresis gel.	1 per lab (optional)	Thermo Fisher Qubit (Q33238). Thermo Fisher link	~\$4,600
Equipment	Nanodrop		For DNA quantification. Similar considerations as the Qubit.	1 per lab (optional)	Thermo Fisher Nanodrop (NDLPLUSGL). Thermo Fisher link	~\$7,200

Nanopore Sequencing Equipment

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Sequencing Equipment	MinION sequencer (M1kC or M1kD)	Yes	Portable device capable of generating up to 50 Gb per run. Note: M1kB model is discontinued.	1 per group or lab	Suitable for a wide range of applications; choose model based on specific needs.	~\$1,000
Sequencing Equipment	Flongle adapter	Likely	Adapter for MinION or GridION ; enables sequencing on Flongle Flow Cells , generating up to 2.8 Gb per run.	1 per group or lab	Ideal for small-scale, cost-effective sequencing projects.	~\$1,460 general only as starter p
Sequencing Equipment	GridION		High-throughput device; can run up to 5 MinION Flow Cells simultaneously, each generating up to 50 Gb per run.	1 per lab (if applicable)	Suitable for labs requiring higher throughput and integrated compute capabilities.	~\$67,000
Sequencing Equipment	P2 Solo		Modular device; can run up to 2 PromethION Flow Cells simultaneously, each capable of generating up to 290 Gb per flow cell. Note: P2 and PromethION flow cells are different and not compatible with MinION flowcells.	1 per lab (if applicable)	Ideal for labs with high-throughput sequencing needs; requires external compute resources.	~\$9,550

**Educator Discounts***

Prices change frequently, and these prices do not necessarily reflect potential educator discounts as they are offered by Oxford Nanopore.

Nanopore Sequencing Kits

Depending on your application, you will need at least one Nanopore [library preparation](#) kit. We feature the following in this guide, see the [Nanopore Store](#) for a full product list.

Category	Item	Essential	Notes	Quantities	Recommendations	Price R
Sequencing Kits	Ligation Sequencing Kit V14 (SQK-LSK114)	Yes	Flexible method for preparing sequencing libraries from dsDNA, including gDNA, cDNA, or amplicons. Involves end-repair, dA-tailing, and adapter ligation .	1 per run (6 genomic samples)	Whole-genome sequencing, amplicon sequencing, and cDNA sequencing.	\$699*
Sequencing Kits	Rapid Barcoding Sequencing Kit 24 V14 (SQK-RBK114.24)	Yes	Enables rapid preparation of up to 24 barcoded sequencing libraries without PCR amplification. Suitable for low- input gDNA.	1 per run (24 barcoded samples)	DNA barcoding, microbial genomics, small genome sequencing, and amplicon sequencing.	\$699*
Sequencing Kits	16S Barcoding Kit 24 V14 (SQK-16S114.24)	Yes	Designed for rapid amplification and sequencing of the 16S rRNA gene, allowing genus-level identification of bacteria. Includes 24 unique barcodes for multiplexing .	1 per run (24 barcoded samples)	Microbiome analysis and bacterial identification.	\$900*



Tip

*Educator Discounts

Prices change frequently, and these prices do not necessarily reflect potential educator discounts as they are offered by Oxford Nanopore.



Tip

Nanopore [Library Prep Kits](#)

We will provide specific recommendations for [Nanopore sequencing kits](#) which have their own specific reagents and associated costs. Additional reagents required by the kit incur additional costs.

Molecular Biology Consumables

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Consumables	Micropipette tips (1-10 µl, 10-100 µl, 100-1000 µl)	Yes	Consider filter tips for PCR and/or with precious samples to avoid cross-contamination.	Multiple boxes per lab	Filter tips recommended for sensitive work.	\$10–\$50
Consumables	0.2 ml PCR tubes	Yes	Essential for PCR workflows.	Multiple packs per lab	High-quality, sterile PCR tubes from trusted brands.	\$10–\$30
Consumables	1.5 ml or 1.7 ml microfuge tubes	Yes	Sterile and DNA/DNase-free strongly recommended.	Multiple packs per lab	Brands like Eppendorf or Thermo Fisher offer reliable options.	\$10–\$30
Consumables	1.7 ml lo-bind tubes		Many Nanopore protocols call for these. Strongly recommended when working with precious samples.	Multiple packs per lab	Eppendorf LoBind Tubes	~\$50 per pack
Consumables	Freezer-safe tape and labels	Yes	For labeling samples stored at -20°C or lower.	Multiple rolls per lab	Freezer-safe tape and lab-specific labels ensure sample integrity.	\$5–\$20
Consumables	Permanent markers	Yes	Useful for labeling tubes and containers.	Multiple per lab	Choose markers with fade-resistant, waterproof ink.	\$1–\$5 per set

Molecular Biology Reagents

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Reagents	Agarose powder	Likely	For preparing electrophoresis gels.	Multiple packs per lab	Molecular biology grade agarose is recommended.	\$50+ per pack
Reagents	DNA loading dye	Likely	Required for visualizing DNA on electrophoresis gels.	1–2 bottles per lab	Choose dyes compatible with your gel documentation system.	Varies, inexpensive
Reagents	DNA stain	Likely	For staining DNA in gels. Options include SYBR Safe, GelRed, or ethidium bromide (use cautiously).	1–2 bottles per lab	Invitrogen SYBR Safe is non-toxic; GelRed is safer than ethidium bromide and has good visualization properties.	\$100–\$200 per bottle
Reagents	Buffer solutions (e.g., TAE or TBE)	Likely	For electrophoresis and sample preparation.	Multiple bottles per lab	Purchase premade buffers or prepare stock solutions.	\$20–\$50 per bottle
Reagents	DNA standards for electrophoresis	Yes	For estimating DNA fragment sizes during gel electrophoresis.	1–2 vials per lab	Use molecular weight ladders appropriate for your protocol, such as 1 kb or 100 bp ladders.	\$20–\$100 per vial
Reagents	Nuclease-free water	Yes	For preparing samples and dilutions.	Multiple bottles per lab	Molecular biology grade water is recommended for all DNA-related protocols.	\$50+ per bottle, depending on quantity
Reagents	200 proof ethanol	Yes	Used in DNA/RNA purification and sample cleanup protocols.	Multiple bottles per lab	Molecular biology grade ethanol ensures no contamination in sensitive workflows.	\$50+ per bottle, depending on quantity
Reagents	Isopropanol	Yes	Commonly used in DNA/RNA precipitation protocols.	Multiple bottles per lab	Molecular biology grade isopropanol is preferred for reliable results.	\$50+ per bottle, depending on quantity

**Tip****DNA Extraction Kits**

We will provide specific recommendations for [DNA extraction kits and protocols](#) which have their own specific reagents and associated costs.

Computers, accessories, software, and online accounts

Category	Item	Essential	Notes	Quantities	Recommendations	Price Range
Computers and Accessories	Laptop or desktop computer*	Yes	Required for running sequencing software and data analysis tools.	1 per lab and/or per sequencing device	Mid-range or higher specifications; 16 GB RAM, quad-core processor recommended.	\$1,000–\$2,000
Computers and Accessories	External storage		Useful for storing large sequencing data files.	Several hundred GB to 1TB+ depending on use	Portable SSDs with fast read/write speeds are ideal.	\$100–\$500
Computers and Accessories	Internet connection	Yes	Required for downloading software updates and accessing cloud-based tools.	1 computer	Stable broadband is sufficient; Wi-Fi or Ethernet compatible.	Varies
Software	MinKNOW	Yes	Primary software for operating Oxford Nanopore sequencing devices.	1 per computer	Comes with Oxford Nanopore devices; ensure latest version is installed.	Free with device
Software	EPI2ME	Likely	Cloud-based data analysis platform for processing sequencing data.	1 installation per computer	Useful for beginners or labs without bioinformatics expertise.	Free or subscription
Online Accounts	Oxford Nanopore account	Yes	Required for purchasing devices, kits, and accessing resources.	1 per instructor	Sign up through the Oxford Nanopore website.	Free
Online Accounts	Cloud storage (e.g., Google Drive, Dropbox)		Useful for sharing and backing up sequencing data. Must have an allocation manager who can share access to their resources.	1 per lab	Choose a storage platform with adequate space for large datasets.	Free or subscription
Online Accounts	NSF ACCESS-CI		National computing infrastructure for advanced data processing.	Depends on configuration	Apply through ACCESS-CI for computational resources.	Free
Online Accounts	JetStream2		Cloud computing resource for scientific workflows. Requires an NSF ACCESS-CI account and allocation for at least one account manager; students	1 allocation per instructor (can be used across multiple students/sections)	Apply through JetStream2 for access to cloud-based computation.	Free

Category	Item	Essential	Notes	Quantities	Recommendations	Price R
			will not necessarily need their own account.			
Online Accounts	DNA Subway		Educational platform for bioinformatics and genomics analysis.	1 account per student, instructor, lab aide	Sign up at DNA Subway for classroom-friendly analysis tools.	Free
Online Accounts	CyVerse		Cloud-based platform for managing and analyzing large biological datasets.	1 account per student, instructor, lab aide	Apply at CyVerse for access to computational and data management tools.	Free or subscrip
Online Accounts	Galaxy		Open-source platform for bioinformatics workflows, accessible via a web interface.	1 account per student, instructor, lab aide	Perform bioinformatics tasks without the need for local installations using Galaxy.	Free



Choosing a Computer to Run a MinION Device*

Selecting the right computer for your MinION device is crucial for smooth sequencing and data analysis. The requirements vary slightly depending on the model you're using and the operating system:

For the MinION M1kB

- **Operating System:** Windows 10 or 11, macOS Monterey (12), Ventura (13), Sonoma (14), or Ubuntu 20.04 or 22.04.
- **Processor:** Intel or AMD processor with at least 4 cores/8 threads.
- **Memory:** 8 GB RAM (minimum).
- **Storage:** 500 GB SSD or higher for managing sequencing data.
- **USB Ports:** Requires USB 3.0 port.

For the full IT requirements, refer to the [MinION M1kB specifications](#).

For the MinION M1kD

- **Operating System:** Windows 10 or 11, Ubuntu 20.04 LTS, or macOS 12 (Monterey) or later.
- **Processor:** Intel or AMD processor with at least 4 cores/8 threads.
- **Memory:** 16 GB RAM or more for optimal performance.
- **Storage:** 1 TB SSD or higher to handle large datasets.
- **USB Ports:** Requires USB-C port.

For the full IT requirements, refer to the [MinION M1kD specifications](#).

General Recommendations

- **Portability:** A laptop or MacBook is ideal for portability, especially for fieldwork or classroom use.
- **Performance:** For labs handling high-throughput sequencing, invest in a computer with higher specifications than the minimum.
- **External Storage:** Consider adding an external SSD to manage and back up sequencing data efficiently.

Choosing the right system ensures reliable performance and smooth operation of your MinION device.

Current best pick - MacBook Pro

While any computer meeting the minimum specifications provided by Oxford Nanopore Technologies will work, we currently recommend the **14-inch MacBook Pro** with the M3 chip for its excellent performance and compatibility. This recommendation may be updated as technologies evolve. If you already have access to a computer meeting Nanopore's minimum requirements, it should work just fine.

Recommended Configuration

- **Processor:** Apple M3 chip with 8-core CPU, 10-core GPU, and 16-core Neural Engine.
- **Memory:** 24 GB unified memory.
- **Storage:** 1 TB SSD for managing sequencing data.
- **Ports:** Two Thunderbolt / USB 4 ports, HDMI port, SDXC card slot, headphone jack, and MagSafe 3 port.
- **Operating System:** macOS Sonoma (14) or later.
- **Display:** 14-inch Liquid Retina XDR for clear and detailed data visualization.

Learn more about this MacBook Pro model on the [Apple website](#).

Why We Recommend This MacBook Pro

1. **Performance:** The M3 chip provides robust performance for running sequencing software like MinKNOW and performing bioinformatics analyses.
2. **Portability:** Its compact size and long battery life make it ideal for classroom or fieldwork settings.
3. **Compatibility:** Fully compatible with Oxford Nanopore devices and software, including macOS-supported versions.
4. **Future-Proofing:** With cutting-edge technology, this MacBook is well-suited for evolving software and sequencing demands.

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3. Equipment and Lab Management

3.1 Oxford Nanopore Sequencing Hardware

3.1.1 The MinION device

Video: [MinION Introduction](#) 

Nanopore documentation: [Hardware](#)

The MinION is a portable, USB-powered device designed for DNA (or RNA) sequencing. At the core of its operation are flow cells, which serve as the functional units where sequencing occurs. Flow cells are consumable cartridges embedded with thousands of nanopores through which DNA or RNA strands pass during sequencing. These flow cells are where the sequencing is happening on the device. The MinION contains the electronics which run the flow cell, keep it at the appropriate temperature, and manage data transfer.



Mk1B Sequencer



Mk1D Sequencer

Technical Summary (Mk1B) - **Technology:** Nanopore sequencing with flow cells - **Flow Cells:** Consumable units containing nanopores; available in various models to support different throughput needs - **Sequencing Output:** Up to 50 Gb per run with a standard MinION flow cell - **Size:** 105 mm x 23 mm x 16 mm - **Weight:** ~87 grams - **Power Source:** USB connection to a laptop or desktop computer - **Software:** MinKNOW for sequencing operation and basecalling; optional EPI2ME for additional analysis

MinIONs - the 'workhorse' for educational sequencing

Generally, we suggest most classroom educators will focus on MinIONs, especially as they get started with sequencing. There are other larger Nanopore sequencing platforms which offer greater data production capabilities but at a higher cost.

MinION device models: Mk1B, Mk1C, and Mk1D

Oxford Nanopore Technologies offers several models of the [MinION](#) device, each tailored to specific sequencing needs.

MinION Mk1B:

The Mk1B is a portable, USB-powered device designed for DNA and RNA sequencing. It connects to a computer via a single USB 3.0 port, providing power, temperature control, and data transfer. The device is compatible with [MinION flow cells](#) and [Flongle](#) adapters, enabling flexibility in sequencing applications.

- [Technical Specifications](#)

MinION Mk1C (Discontinued):

The Mk1C integrates sequencing hardware with built-in computational capabilities, offering a compact and portable solution for nanopore sequencing. This self-contained device eliminates the need for external computing resources, as it combines sequencing and analysis within a single unit. This device is discontinued and although still functional, at some point updates to the software may be discontinued. If you have questions about this device you should contact Nanopore [support](#) directly.

- [Technical Specifications](#)

MinION Mk1D:

The Mk1D features improved thermal dissipation capabilities, enhancing sequencing performance and enabling accurate real-time sequencing across diverse environments. It includes intuitive status LEDs for easy monitoring of device status. The Mk1D is compatible with [MinION flow cells](#) and [Flongle](#) adapters, supporting both DNA and RNA sequencing.

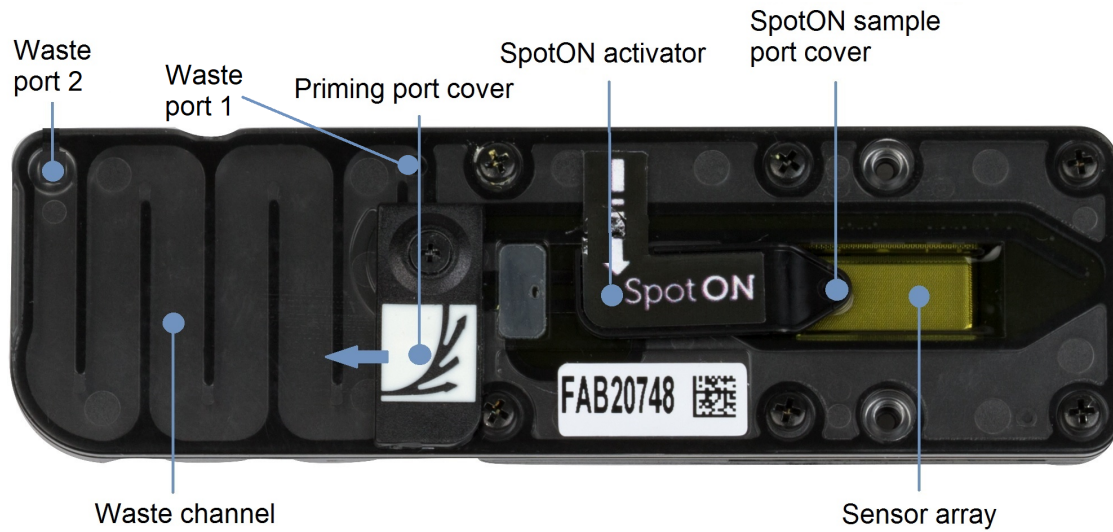
- [Technical Specifications](#)

3.1.2 Flow cells

[Flow cells](#) are essential components of Nanopore sequencing devices. Each [flow cell](#) contains a membrane embedded with an array of nanopores. These nanopores form the core of the sequencing process by detecting the passage of nucleic acid strands through the [flow cell](#). When DNA or RNA molecules pass through a nanopore, they cause specific disruptions in an electrical current. These disruptions are recorded and analyzed to determine the sequence of nucleotides. This real-time process enables users to generate sequencing data as the experiment progresses.

Flow cells are available in different designs to accommodate varying experimental needs.

- **MinION Flow Cells:** These flow cells can generate up to 50 gigabases (Gb) of data in a single run, making them suitable for high-throughput applications.



MinION flow cell

- **Flongle Flow Cells:** These are designed for smaller-scale experiments, capable of producing up to 2.8 Gb of data, offering a cost-effective option for pilot studies or limited sample sets. We recommend most educational applications strongly consider Flongles due to their lower cost (<\$70 per flow cell).

Adapter required

Flongle flow cells come with an adapter that allows them to fit into the MinION devices (see figure below).



Flongle flow cell (above) and Flongle adapter (below)

Flow cell reminders

- **Inspect before use:** Visually inspect flow cells upon receipt and before use to ensure no visible damage or contamination. Report any issues immediately to Nanopore support.
- **Run flow cell check before loading:** Before you load your sequencing library, you can use MinKNOW software to check your flow cell and ensure it is functioning and within warranty. Contact Nanopore support for help if a flow cell does not pass. Keep in mind old or expired flow cells may still yield good-enough results, but don't depend on this when planning.
- **Maintain records:** Keep records of purchase dates, storage conditions, and expiration dates.
- **Plan shipping and use promptly:** Plan experiments to use flow cells as close to their purchase date as possible to maximize reliability. You can generally plan (or adjust) the ship date of your Nanopore reagents and flow cells to arrive as needed within 1 year of purchase. If you order multiple flow cells you can even break up shipments. Keep in mind that each shipment may incur its own shipping charges.

**Don't freeze your flow cells - and other tips****Proper Storage and Handling of Flow Cells**

Flow cells are delicate, high-precision consumables that require specific storage and handling conditions to maintain their performance. Improper storage or use can result in reduced sequencing quality or failure.

Storage and temperature requirements

- **Storage Temperature:** Flow cells must be stored at 2°C to 8°C (refrigerated) to preserve the integrity of the nanopores and other components. They should never be frozen or exposed to temperatures outside this range.
- **Ensure that whomever will receive your Nanopore shipments are aware of this restriction.** Your shipment may contain reagents that require freezing, and it is not uncommon for an unsuspecting person in receiving or a lab manager inexperienced with Nanopore to accidentally freeze flow cells.

Shelf Life

See [Flow cell warranty and storage](#)

- Flow cells have a limited shelf life, and their performance is guaranteed only within the expiration date provided by Oxford Nanopore. Always check the expiration date before use and prioritize using older flow cells first to minimize waste.
- **MinION shelf life:** ~12 weeks
- **Flongle shelf life:** ~4 weeks
- Flow cells come with a warranty that covers manufacturing defects. However, this warranty does not apply if flow cells are damaged due to improper storage, handling, or use. Always follow the manufacturer's guidelines to ensure eligibility for warranty claims.

3.1.3 Other Nanopore sequencing platforms

Video: [Choosing the right Nanopore sequencing device for you](#) 

Oxford Nanopore offers a range of sequencing platforms beyond the [MinION](#), designed for higher-throughput and more specialized applications. While these platforms provide significant sequencing capacity and flexibility, their cost and complexity make them better suited for dedicated research projects or advanced educational applications. For most classrooms and undergraduate settings, the [MinION](#) platform remains the preferred choice due to its affordability, portability, and ease of use.

GridION The GridION is a benchtop sequencing platform that can operate up to five [MinION](#) flow cells simultaneously, allowing for increased throughput. It includes integrated compute capability, enabling data analysis alongside sequencing. This device is ideal for labs requiring moderate to high-throughput sequencing in a compact setup.

- [Learn more about the GridION](#)

PromethION and P2 Solo The PromethION series includes high-throughput sequencing platforms capable of handling up to 48 flow cells simultaneously in its larger models. The PromethION P2 Solo is a more compact option, supporting two PromethION flow cells, each capable of generating up to 290 gigabases (Gb) of data. These devices are suited for large-scale projects, such as sequencing entire genomes or extensive metagenomic studies.

- [Learn more about the PromethION series](#)

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3.2 Library Preparation and Sequencing Kits

3.2.1 Core Nanopore sequencing kits

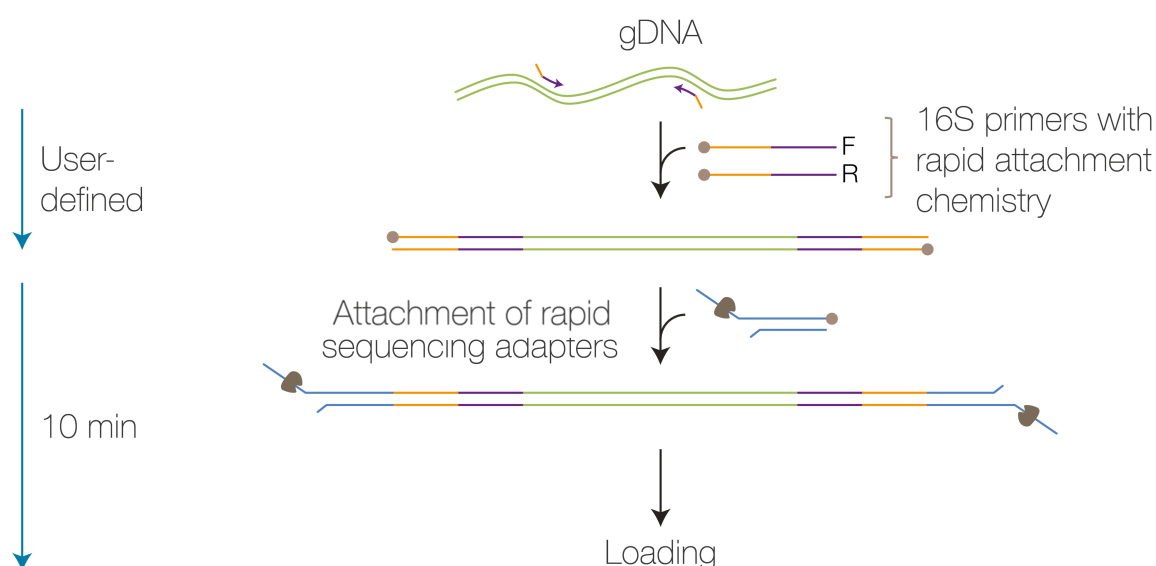
While Oxford Nanopore offers a variety of kits and applications, we generally focus on sequencing strategies that use the following three Nanopore kits.

Nanopore account required to access

Some of the links require you to sign into your Oxford Nanopore account. You may obtain a free account by signing up: [Register](#)

16S Barcoding Kit 24 V14

The **16S Barcoding Kit 24 V14** is designed for rapid and efficient amplification and sequencing of the full-length 16S rRNA gene (~1,500 bp) from extracted genomic DNA (gDNA). This kit facilitates genus-level bacterial identification by targeting the 16S rRNA gene, a highly conserved region present in all bacteria, making it a valuable tool for microbial diversity studies.



16s Workflow (as presented on Oxford Nanopore Product Page)

Key Features

- **Full-length amplification:** Utilizes primers that amplify the entire 16S rRNA gene for accurate bacterial identification.
- **Multiplexing:** Includes 24 unique barcodes, allowing simultaneous sequencing of up to 24 different samples in a single run, enhancing throughput and reducing costs.
- **Rapid library preparation:** The protocol requires approximately 40 minutes plus PCR time (2-3 hours).
- **Minimal input requirement:** Requires only 10 ng of gDNA per sample, making it suitable for samples with limited DNA quantities.
- **No fragmentation:** Designed to work with intact gDNA without the need for fragmentation, simplifying the preparation process.

Applications

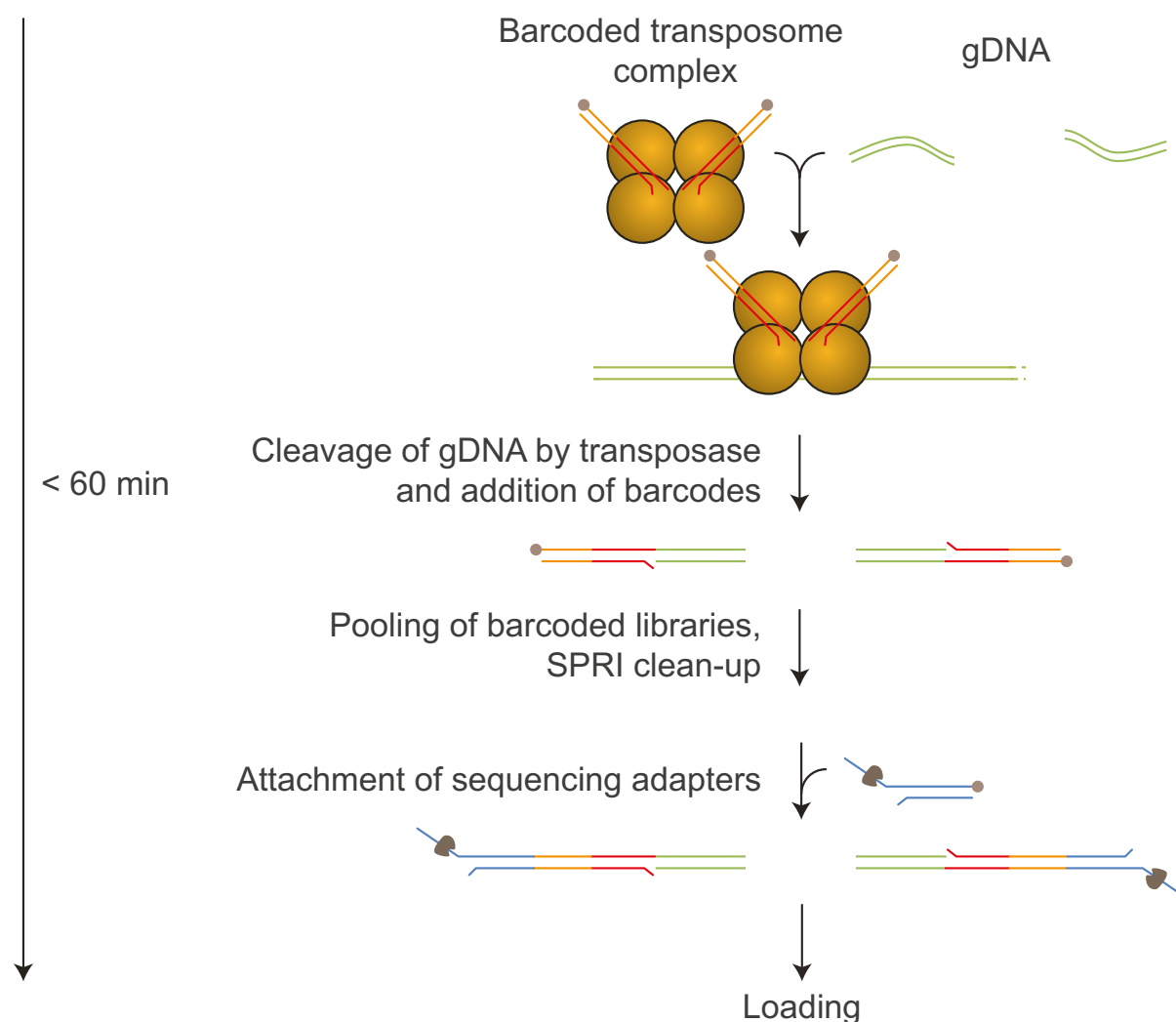
- **Microbial community profiling:** Ideal for analyzing complex microbial populations in environmental samples, or microbiome studies.
- **Bacterial identification:** Enables rapid identification of bacterial genera in mixed samples, aiding in diagnostics and research.
- **Educational applications:** Suitable for teaching laboratories to demonstrate sequencing techniques and microbial diversity analysis.

What you get (and what's not included)

The product page for all Nanopore kits has a "[3RD Party Materials](#)" section that describes all of the materials and reagents **not included** in the kit. Review this list ahead of time to ensure you have everything you need to perform the experiment.

Rapid Barcoding Kit 24 V14

The [Rapid Barcoding Kit 24 V14 \(SQK-RBK114.24\)](#) is designed for swift and straightforward preparation of up to 24 barcoded sequencing libraries without the need for PCR amplification. This kit employs a transposase enzyme to simultaneously fragment genomic DNA (gDNA) and attach unique barcoded tags, facilitating efficient multiplexing and sequencing.



Example rapid sequencing kit protocol (as presented on Oxford Nanopore Product Page). Note: Your protocol may vary based upon application.

Key Features:

- **PCR-Free workflow:** Enables direct sequencing of gDNA without PCR amplification, preserving base modifications and reducing preparation time.
- **Multiplexing:** Includes 24 unique barcodes, allowing simultaneous sequencing of up to 24 samples in a single run, enhancing throughput and cost-effectiveness.
- **Rapid library preparation:** The protocol requires approximately 60 minutes, streamlining the workflow for quick results.
- **Minimal input:** Requires only 200 ng of gDNA per sample, accommodating samples with limited DNA quantities.

To PCR or not to PCR

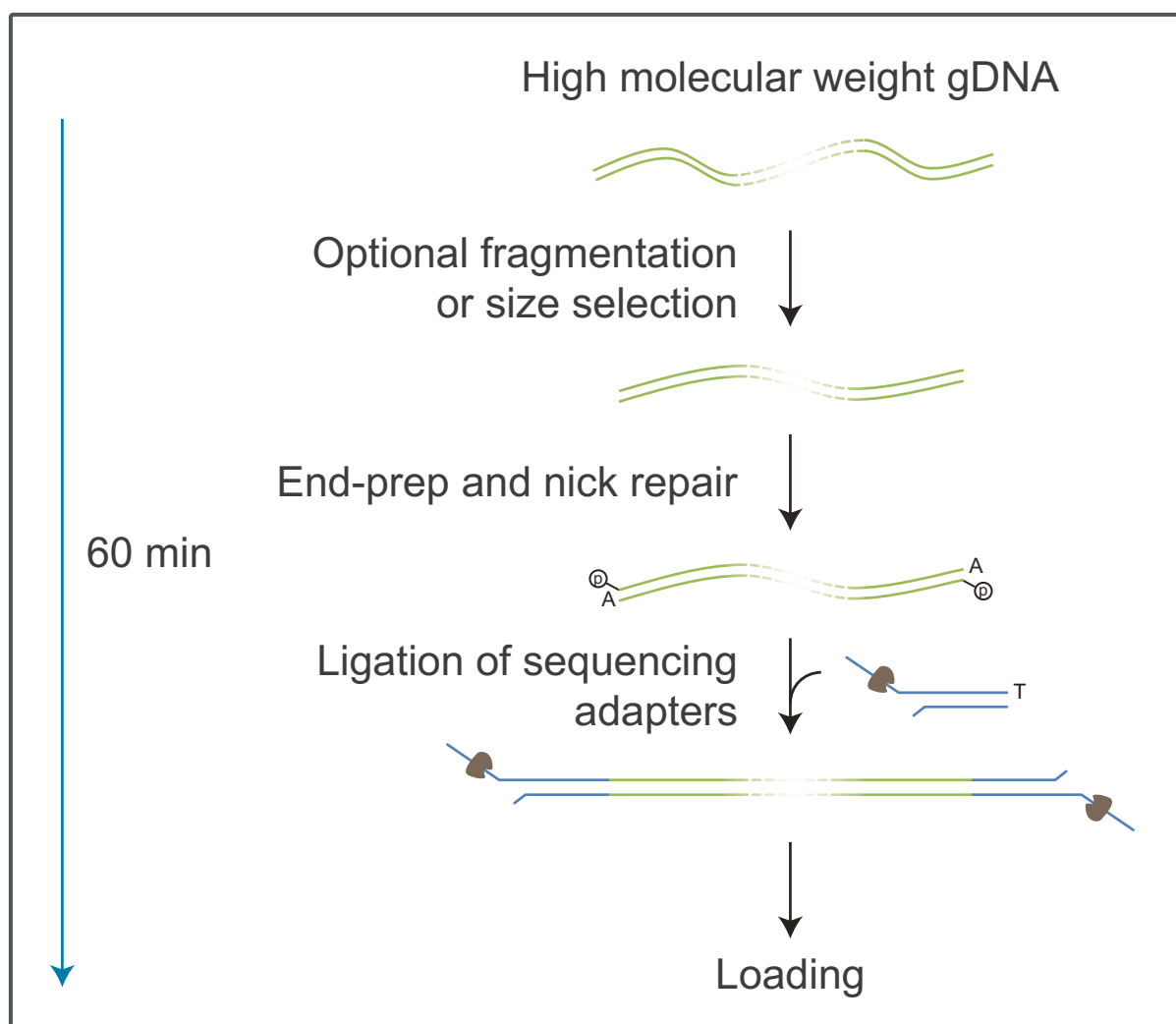
We also recommend this kit for DNA Barcoding (e.g. species identification using DNA); in that case, you will amplify a barcoding region (e.g., *rbcL*, *COI*) and using this kit to sequence the amplicon.

Applications:

- **DNA Barcoding:** Barcode regions amplified by PCR or other amplicons can quickly be sequenced using this kit. Because this kit fragments PCR products, you will need to use bioinformatics to reassemble fragments.
- **Genomic DNA sequencing:** Suitable for rapid sequencing of gDNA from various organisms, aiding in genomic studies and research.
- **Microbial community analysis:** Facilitates the study of complex microbial populations by enabling the sequencing of multiple samples concurrently.

Ligation Sequencing Kit V14

The [Ligation Sequencing Kit V14](#) offers a flexible method for preparing sequencing libraries from double-stranded DNA (dsDNA), including genomic DNA (gDNA), complementary DNA (cDNA), or amplicons. The straightforward [library preparation](#) involves repairing DNA ends and adding a dA-tail using the NEBNext End Repair/dA-Tailing Module, followed by [ligation of sequencing adapters](#) provided in the kit.



Ligation sequencing kit protocol (as presented on Oxford Nanopore Product Page).

Key Features:

- **Versatile input compatibility:** Suitable for various dsDNA inputs, such as gDNA, cDNA, or amplicons, providing flexibility across different applications.
- **High accuracy:** Optimized to deliver modal raw read accuracies over 99% (Q20+) when used with R10.4.1 nanopores, ensuring reliable sequencing data.
- **Long read lengths:** Capable of producing read lengths equivalent to the input fragment length, facilitating comprehensive genomic analyses.
- **Preparation time:** Approximately 120 minutes, streamlining the workflow for efficient library preparation.
- **Input requirement:** Requires 1,000 ng of dsDNA; for fragmented or PCR-amplified DNA, 100 ng or more is sufficient, and for amplicons or cDNA, 100–200 fmol is recommended.

Applications:

- **Whole genome sequencing:** Ideal for sequencing entire genomes, providing comprehensive coverage and high accuracy.
- **Transcriptome analysis:** Suitable for sequencing cDNA to study gene expression and alternative splicing events.
- **Amplicon sequencing:** Effective for sequencing PCR products, enabling targeted analysis of specific genomic regions.

3.2.2 Additional sequencing reagents

The [sample prep](#) product page features the entire line of kits. Some additional ones you may want to be aware of include:

- **Native barcoding Kit:** Similar to the ligation kit, allows you to barcode your ligated products (e.g., genomic DNA, amplicons)
- **Flow cell wash kit:** If you are using the MinION flow cell, you can use this kit to wash between uses, eliminating cross-contaminating DNA.
- **Sequencing auxiliary vials:** During your trials or if you are doing lots of sequencing, you may eventually find you have run out of some reagents and have extra of others. These are extra vials of common reagents, allowing you to have extra on hand.
- **Rapid adapter auxiliary:** If you are using kits with the rapid adaptor (e.g. 16s, rapid barcoding), this is the most expensive component and a limiting reagent. If you want extra, you may be able to pool and extend your kits.

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3.3 Computer and Software Management for Nanopore Sequencing

3.3.1 Computer Requirements

To operate a Nanopore sequencing device, selecting the right computer and software is essential for smooth sequencing runs, data management, and analysis.

The Role of Computers in Nanopore Sequencing Workflows

Computers support every stage of the sequencing process, and their use can be divided into four general functions:

1. To operate and control the sequencing device

The computer is responsible for running the software (e.g., [MinKNOW](#)) that controls the sequencing device and monitors the sequencing process in real time. This task does not require a high-end machine; a laptop or desktop with a quad-core processor and 16 GB of RAM is sufficient.

2. To provide storage for sequencing data

Sequencing experiments can generate substantial amounts of data, even with small-scale experiments producing several gigabytes. As the number and complexity of experiments grow, so do storage needs. Solid-state drives (SSDs) are recommended due to their faster read/write speeds, which are crucial for smooth operation during sequencing runs. Ample storage space (1 TB or more) ensures you can manage and retain large datasets over time but a few hundred GB should be sufficient for small-scale applications.

3. To perform basecalling

[Basecalling](#) is the computational process that converts raw electrical signals from the sequencer into nucleotide sequences. While this can be done on most modern computers, a well-equipped machine with a discrete [GPU](#) significantly improves processing speed and accuracy. GPUs enable the use of advanced [basecalling](#) algorithms that enhance sequence quality, particularly for high-throughput applications.

4. To conduct data analysis

Data analysis involves tasks such as genome assembly, variant calling, and classification of sequencing reads. While these can require substantial computational resources, many free cloud-based platforms (e.g., [EPI2ME](#), [Galaxy](#), or [CyVerse](#)) reduce the need for high-end hardware. As a result, lower-specification computers can still be used effectively for data analysis, particularly when combined with cloud resources.

GPUs and basecalling

What is basecalling?

[Basecalling](#) is the process of converting raw electrical signals generated during sequencing into nucleotide sequences (A, T, G, C). High-quality [basecalling](#) enhances the accuracy of downstream analyses.

Using GPUs for basecalling

While [basecalling](#) can be performed without a [GPU](#), access to GPUs significantly accelerates the process and supports the highest-quality [basecalling](#) algorithms. Modern discrete GPUs, such as NVIDIA GeForce RTX or equivalent, improve performance dramatically.

Accessing GPUs

- **JetStream2 and ACCESS-CI:** Free access to [GPU](#) resources is available through these academic platforms. While powerful, they require some technical expertise to set up and use effectively. [Learn more about JetStream2](#).
- **Amazon Web Services (AWS):** Provides scalable [GPU](#) resources for [basecalling](#) and data analysis. However, these services incur hourly costs, depending on the instance type.

No GPU? No problem!

If you do not have access to a [GPU](#), [MinKNOW](#) includes CPU-based [basecalling](#) that is sufficient for many educational and smaller-scale projects. However, processing times will be longer, and throughput may be reduced.

Matching computer resources to workflow needs

The computer used to run the [MinION](#) can be a high-performance machine to manage device operation, data storage, and [basecalling](#) efficiently. However, student computers used for limited or cloud-based data analysis tasks don't need to meet such demanding requirements. Cloud platforms provide cost-effective ways to leverage advanced computational tools without requiring expensive hardware, ensuring accessibility for educators and students.

Minimum System Requirements

Oxford Nanopore devices, such as the [MinION Mk1B](#) and [Mk1D](#), have specific requirements to ensure reliable operation. At a minimum, your computer should meet the following specifications:

- **Operating system:**
 - Windows 10 or later (64-bit)
 - macOS 12 Monterey or later (Apple M2/M3/M4 series or Intel processors)
- **Processor:**
 - Quad-core CPU (Intel i5/i7 or AMD equivalent) or Apple M2/M3/M4 series.
- **RAM:**
 - At least 16 GB for efficient operation
- **Storage:**
 - 500 GB SSD for data handling and storage
- **USB Ports:**
 - One USB 3.0 port for device connection
- **Graphics:**
 - Integrated graphics are sufficient for basic operations; discrete GPUs recommended for intensive tasks (see below for details on [GPU use](#)).

 **Warning: Do your homework - computer specifications change frequently**

Please check [MinION IT requirements](#) or contact [Nanopore support](#) before making a purchase. This guide is not a replacement for doing your own homework!

Recommended Systems

For optimal performance, especially when handling large datasets or performing local [basecalling](#), consider systems with higher specifications. Modern laptops such as the Apple MacBook Pro (M2/M3/M4 series) or high-performance Windows laptops meet these needs. The following have appeared on Nanopore educational resources as potential choices.

- Apple MacBook Pro: [Retail site](#)
- Razer Blade 18: [Retail site](#)

3.3.2 Data Storage in Nanopore Sequencing Workflows

Data storage is a critical aspect of Nanopore sequencing workflows, as even small experiments can produce several gigabytes (GB) of data. Proper storage solutions are essential for managing this data efficiently and ensuring it is secure and accessible for analysis and archiving. Both external solid-state drives (SSDs) and cloud-based storage options are options, each with advantages and limitations. Investing in appropriate storage solutions and a solid backup strategy, educators and researchers can effectively manage their sequencing data, ensuring its security, accessibility, and readiness for downstream analysis or publication.

External Solid-State Drives (SSDs)

External SSDs offer fast, reliable, and portable storage solutions, making them an excellent choice for sequencing workflows. The high read/write speeds of SSDs are especially important during sequencing runs, where real-time data transfer is required. Their durability and portability make them suitable for laboratory and fieldwork applications.

Recommendations for external SSDs

- **Capacity:** A minimum of 500 GB is recommended for basic projects, but 1 TB or larger is ideal, especially for labs handling multiple datasets or larger-scale experiments.
- **Performance:** Drives with high read/write speeds, such as NVMe or USB 3.2, are highly recommended to ensure efficient data handling during sequencing and analysis.

Cloud-Based storage solutions Cloud storage provides scalable and convenient options for managing sequencing data, particularly for collaborative or remote projects. Platforms like Google Drive, Dropbox, and AWS S3 allow users to upload and share data without relying on physical storage devices.

Considerations for cloud storage

- **Capacity and costs:** While many platforms offer free storage tiers, these often have file size limits that can be restrictive for large sequencing datasets. Paid plans with higher storage capacities may be necessary as data volume increases.
- **Accessibility:** Cloud solutions facilitate collaboration by enabling remote access and data sharing with colleagues or collaborators.
- **Backup:** Cloud platforms can serve as a secondary or backup solution to complement local storage, ensuring data redundancy.

Data backup and archiving

For important datasets, particularly those intended for publication, a robust data backup strategy is essential. Backing up data on multiple devices or platforms minimizes the risk of data loss due to hardware failure or accidental deletion. Consider the following practices:

- Use external SSDs for local backups and a cloud storage platform for off-site redundancy.
- Regularly verify that backups are complete and accessible.
- Implement a naming and organization strategy to ensure datasets are easy to locate and reference.

Important caveats and practices

While external and cloud-based storage solutions are effective, users should also consider:

- **Data privacy:** Sensitive datasets, especially in clinical or ecological research, must comply with data privacy regulations.
- **FAIR Data management:** Adhering to FAIR (Findable, Accessible, Interoperable, Reusable) principles is encouraged to promote data stewardship and reuse.

3.3.3 Software Requirements

Video: [How to run your sequencing device and get started with data analysis](#) 

At a minimum, [MinKNOW](#) and [EPI2ME](#) software will be needed for Nanopore sequencing. In the [bioinformatics](#) section of this guide, we will provide and review additional information.

[MinKNOW](#)

[MinKNOW](#) is the primary software for operating Nanopore sequencing devices. It controls the sequencing hardware, monitors flow cell performance, and performs real-time basecalling.

- **Download:** Available after registering an Oxford Nanopore account.
- **Compatibility:** Pre-installed on [MinION Mk1C](#) and [Mk1D](#); compatible with Windows and macOS systems for [Mk1B](#) and [Mk1D](#).

[EPI2ME](#)

[EPI2ME](#) is a cloud-based platform for data analysis. It complements [MinKNOW](#) by providing tools for downstream tasks, such as read classification and genome assembly. [EPI2ME](#) is suitable for users with limited local computing resources.

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4. Laboratory Protocols

4.1 Sample Collection

4.1.1 DNA Sampling and Extraction for Nanopore Sequencing

Introduction

Nanopore sequencing is a highly versatile technology capable of analyzing DNA directly from diverse sample types, from environmental sources to purified genomic extracts. However, the success of a sequencing experiment begins with DNA sampling and extraction—ensuring that high-quality DNA is obtained while keeping protocols accessible and cost-effective.

In this guide, we focus on DNA extraction methods suitable for classroom and teaching labs where simplicity, affordability, and safety are key considerations. We generally recommend approaches that:

- **Minimize toxic chemicals** to ensure safety and ease of disposal.
- **Reduce reliance on specialized equipment** to keep protocols accessible.
- **Lower costs** by recommending practical and scalable methods.

In many cases, particularly when PCR amplification is used, simpler homemade DNA extractions can be sufficient, even when DNA integrity is not perfect. For applications that require long DNA reads such as whole-genome sequencing or structural variant analysis, we will highlight commercial extraction kits that give a better chance for high DNA integrity.

While specialized extraction methods may yield the highest-quality DNA, they often require additional steps, equipment, and reagents.

Basic principles of DNA extraction

PubMed lists more than [1600 articles](#) on DNA extraction methods. Depending on your sample, you can find specialized methods, but we focus here on the most generalizable approaches knowing that many educators deploying in the classroom will want to focus on 1-2 methods that can cover the largest variety of samples. DNA extraction usually involves the following:

Cell lysis

A lysis buffer containing detergents, mechanical disruption, and heat are all commonly used tools. Mechanical disruption methods, such as bead beating, grinding with a mortar and pestle, or using liquid nitrogen for tough samples like plant tissue, help break open cell and nuclear membranes, allowing cellular material to be released into solution.

Protein digestion and removal

Once the cells are lysed, enzymes such as proteinase K are often added to degrade proteins that might be bound to the DNA or otherwise contaminate the sample. In addition to enzymatic digestion (e.g., lysozyme for bacterial samples), chemical agents (e.g., phenol or chloroform) can be used to separate proteins from the nucleic acids. This step helps ensure that the final DNA preparation is as free as possible from protein contaminants. We don't recommend any methods using phenol due to the increased safety hazard.

Precipitation

To isolate the DNA, an alcohol (usually ethanol or isopropanol) is added along with salts. The salts help neutralize the charges on the DNA, and the alcohol causes the DNA to precipitate out of solution. Centrifugation is then used to collect the DNA as a pellet. Most protocols will use isopropanol first since lower volumes are more efficient, and ethanol is used in washing since it has a lower evaporation point. Two important methods worth highlighting here:

- **Silica membrane-based purification:** DNA binds to a silica column in the presence of chaotropic salts and is later eluted in water or buffer. Many commercial kits use this method and although spin columns add to the cost, column-based kits are fairly reliable and reproducible. A potential drawback is that columns may yield more fragmented DNA (i.e., DNA with lengths in the 1000's of basepairs rather than 10,000-100,000+ bp)
- **Magnetic bead purification:** DNA attaches to magnetic beads, allowing for easy separation and cleanup. This method will frequently be used in Nanopore kits at various cleanup steps to purify DNA after enzymatic steps, rather than the initial DNA extraction.

Washing and Resuspension

The DNA pellet is washed (often with ethanol) to remove any residual salts or other impurities. After washing, the pellet is dried and then resuspended in a suitable buffer (commonly TE buffer) to stabilize the DNA for storage and subsequent applications.



Additional Reading

The Evolution of DNA Extraction Methods. Preetha J Shetty, The Evolution of DNA Extraction Methods, 2020 - 8(1). AJBSR.MS.ID.001234. DOI:[10.34297/AJBSR.2020.08.001234](https://doi.org/10.34297/AJBSR.2020.08.001234).

DNA extraction recommendations by sample type and analysis endpoint

These are some selections we have tried and do not represent all possible or even the best possible approaches. Please give us feedback on new and/or lower-cost approaches you wish to share!

Note: All prices are estimates and may not include the total sample cost, for example including reagents or supplies not provided and/or potential education discounts.

PHAGE

A rapid competitive method for bacteriophage genomic DNA extraction

- **Format:** Homemade
- **Sequencing endpoint:** PCR amplification/16S
- **Costs:** Low (<\$1 sample)
- **Application notes:** [Protocol](#)

Norgen Phage DNA Isolation

- **Format:** Commercial kit
- **Sequencing endpoint:** Whole genome
- **Costs:** Midrange (\$5 sample)
- **Application notes:** [Website](#)

See **Monarch® HMW DNA Extraction Kit for Tissue** Phage protocol

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

BACTERIA

See **Monarch® HMW DNA Extraction Kit for Tissue** Bacteria protocol

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome or PCR amplification/16S
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

Swab collection and DNA preservation

- **Format:** Commercial kit
- **Sequencing endpoint:** DNA collection and preservation only (no DNA extraction)
- **Costs:** Midrange (\$7 per sample + kit for DNA extraction)
- **Application notes:** [Website](#)

Alkaline Method for Bacterial DNA Extraction

- **Format:** Homemade
- **Sequencing endpoint:** PCR amplification/16S
- **Costs:** Low (<\$1 sample)
- **Application notes:** [Protocol](#)

GENERAL MICROBIAL

Norgen Microbiome DNA Isolation Kit

- **Format:** Commercial kit
- **Sequencing endpoint:** PCR amplification/16S
- **Costs:** Midrange (\$6 per sample)
- **Application notes:** [Website](#)

SOIL, ENVIRONMENTAL

DNeasy PowerSoil Pro Kit

- **Format:** Commercial kit
- **Sequencing endpoint:** PCR amplification/16S, whole genome
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

WATER, ENVIRONMENTAL

E.Z.N.A.® Water DNA Kit

- **Format:** Commercial kit
- **Sequencing endpoint:** PCR amplification/16S
- **Costs:** Low (\$4 per sample)
- **Application notes:** [Website](#)

PLANT

See **PacBio Nanobind PanDNA kit** formatted for plant nuclei lysis

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$14 per sample)
- **Application notes:** [Website](#)

INSECT

See **Monarch® HMW DNA Extraction Kit for Tissue** Insect protocol

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

GENERAL

Simple, Robust Invertebrate DNA Barcoding: Chelex-Based DNA Extraction and Optimized COI Amplification

- **Format:** Homemade
- **Sequencing endpoint:** PCR amplification
- **Costs:** Low (<\$1 per sample)
- **Application notes:** [Website](#)

Fire Monkey High Molecular Weight DNA (HMW-DNA) extraction kit

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

Monarch® HMW DNA Extraction Kit for Tissue

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$10 per sample)
- **Application notes:** [Website](#)

PacBio Nanobind PanDNA kit

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** High (\$14 per sample)
- **Application notes:** [Website](#)

New England Biolabs® (NEB) support for ORG.one participants

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** Varies
- **Application notes:** [Website](#)

Promega Wizard® HMW DNA Extraction Kit

- **Format:** Commercial kit
- **Sequencing endpoint:** Long-read whole genome
- **Costs:** Midrange (\$7 per sample)
- **Application notes:** [Website](#)

**Additional Reading**

1. Oxford Nanopore **Extraction protocols** page (sign-in required) <https://nanoporetech.com/documentation/results?category=prepare&topic=extraction-protocols>.
2. **DNA Barcoding 101** — DNA extraction protocols. [Website](#)
3. New England Biolabs® (NEB) support for ORG.one participants. <https://nanoporetech.com/oo/extraction-protocols>.
4. Gand, M., Bloemen, B., Vanneste, K. et al. **Comparison of 6 DNA extraction methods for isolation of high yield of high molecular weight DNA suitable for shotgun metagenomics Nanopore sequencing to detect bacteria.** BMC Genomics 24, 438 (2023). <https://doi.org/10.1186/s12864-023-09537-5>.

🕒 February 10, 2025

🕒 February 6, 2025

5. Curated Experiments

5.1 Curated Experiment: DNA Barcoding

5.1.1 Identifying Species Using DNA Barcoding

Summary: In this experiment, DNA extracted from plants, insects, or foods can be used to identify a species.



Downloads

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Protocol information

Protocol Credits

- **Author(s)**
- Anna Feitzinger, Cold Spring Harbor Laboratory
- Jeffry Petracca, Cold Spring Harbor Laboratory
- **Maintainer/contact:** Anna Feitzinger, Cold Spring Harbor Laboratory: [email](#)
- **Last updated:** January, 2025
- **Source materials and references**
- [Barcoding 101 Lab Resources](#)
- [Rapid Barcoding Protocol](#)

DNA sample source

- **Type:** Plant, insect, food
- **Collection source:** Varies

Nanopore Sequencing

- **Sequencing format:** Flongle
- **Sequencing kit:** [Rapid Barcoding Kit \(Rapid Barcoding Kit 24 V14\)](#)
- **Oxford Nanopore Sequencing protocol:** [Rapid Barcoding Kit \(SQK-RBK114.24\)](#)
- **Indexed/Barcoded:** Yes, 24 indices
- **Samples per run:** 24 samples

Computer and Bioinformatics

- **Analysis tools**
- [EPI2ME](#)
- DNA Subway 2.0 (coming soon)
- **Analysis difficulty:** Easier
- **Command line needed:** No
- **GPU/Super-high accuracy basecalling needed:** Not necessary, but recommended

Reagents and Sample Prep

Sample collection and prep

- See resources at [Barcoding101 Sample Collections](#)

DNA extraction

- See the rapid DNA extraction protocol at [Barcoding101 Rapid DNA Extraction](#)

DNA prep, library creation, and sequencing

- Rapid Barcoding Kit (SQK-RBK114.24)
- Flongle Expansion Kit (EXP-FLP002)
- Flongle Flow Cell (FLG-FLO114)

Equipment and Consumables

Lab Equipment

- Micropipettes and tips (2-1000 µL)
- Magnetic microfuge tube racks
- Microcentrifuge
- Thermal cycler
- Qubit™ tubes (optional)

Consumables

- 1.5 mL microfuge tubes
- PCR tubes
- Qubit™ fluorometer, Nanodrop™, QIAxpert®, or other method of DNA quantification (optional)

Nanopore Sequencing Equipment

- MinION Flongle Adapter
- MinION Mk1B sequencer

Computer Equipment

- Desktop or laptop with [MinKNOW](#) software

Estimated timings

Section I

Section II

Section III

Section IV

Section V

5.1.2 Background

DNA amplicons are sections of DNA produced via amplification, often using the polymerase chain reaction (PCR). Amplicons may be sequenced to identify organisms (as in DNA barcoding), detect pathogens, analyze genetic variation, or study inherited traits.

Oxford Nanopore Technologies (ONT) sequencing allows for real-time analysis using the MinION device, which is particularly useful for educational applications. To prepare DNA amplicons for sequencing, a sequencing library must first be created. The ONT Rapid Barcoding Kit simplifies this process using a **transposome complex**—a transposase enzyme and associated DNA molecule—to fragment and modify DNA. These modifications include the addition of **barcodes** (oligonucleotide indices), which uniquely identify each DNA sample.

Once prepared, DNA molecules pass through nanopores embedded in a membrane within a flow cell. The nanopore detects changes in ionic current as nucleotides traverse the pore, and these signals are recorded and later **basecalled** into sequence data.

This protocol walks educators through PCR cleanup, barcode attachment, library pooling, and sequencing using a Flongle Flow Cell (FLG-FLO114), which is the recommended sequencing platform for educational applications.

5.1.3 AMPure XP Solid Phase Reversible Immobilization (SPRI) Bead PCR Clean-Up (Optional)

Summary

This protocol, adapted from Beckman Coulter Life Sciences, is for PCR cleanup using AMPure XP beads. It removes residual primers and polymerase while enriching DNA fragments in the range of 400-800 bp, making it suitable for standard barcode marker regions such as **rbcL** and **COI**.

Steps:

1. Resuspend AMPure XP beads by pipetting or vortexing.
2. Add the entire PCR reaction (~20 µL) to 36 µL of AMPure XP beads in a labeled 1.5 mL microfuge tube. Mix gently.
3. Incubate for 5 minutes at room temperature.
4. Spin down briefly, then place the tube on a magnetic rack for 1 minute.
5. Carefully remove and discard the supernatant without disturbing the pellet.
6. Keeping the tube on the magnet, add 200 µL of 70% ethanol, then remove and discard.
7. Repeat Step 6.
8. Air-dry the pellet for 30 seconds. **Caution:** Do not overdry.
9. Remove the tube from the magnet and resuspend the pellet in 10 µL of **Elution Buffer (EB)**.
10. Incubate for 5 minutes at room temperature.
11. Spin down briefly and place the tube back on the magnet.
12. Transfer 10 µL of the supernatant to a fresh PCR tube. This represents the cleaned amplicon sample, ready for barcode attachment.

Flow Cell Check

Summary

Before sequencing, the number of active nanopores on the flow cell must be determined. We recommend Flongle flow cells for amplicon sequencing, as they contain a maximum of 126 nanopores (compared to 2048 in a MinION flow cell).

Steps:

1. Open MinkNOW software.
2. Click the **Start** tab.
3. Select **Flow Cell Check**.

 important

A flow cell is **under warranty** if tested within **2 weeks** of delivery, stored at **4°C**, but has **<50 pores**. Contact Oxford Nanopore within 2 days if replacement is needed.

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5.2 Curated Experiment: 16S Sequencing

5.2.1 Examining Microbial Diversity — Nanopore "Shoe-ome" Sequencing

Summary: In this experiment, students swab the bottom of their shoes to sequence the microbial diversity of their footwear.



Downloads

Download: Download a copy of this template [In Markdown](#) or [Microsoft Word](#)



Protocol information

Protocol Credits

- **Author(s)**
- Jason Williams, Cold Spring Harbor Laboratory
- **Maintainer/contact:** Jason Williams, Cold Spring Harbor Laboratory: [email](#)
- **Last updated:** January, 2025
- **Source materials and references**
- [Norgen Microbiome Kit](#)
- [Swab Collection and DNA Preservation System](#)

DNA sample source

- **Type:** Microbial
- **Collection source:** Collected from swabs of shoes/footwear

Nanopore Sequencing

- **Sequencing format:** Flongle
- **Sequencing kit:** [16S Barcoding Kit 24 V14 \(SQK-16S114.24\)](#)
- **Oxford Nanopore Sequencing protocol:** [Official ONT protocol](#)
- **Indexed/Barcoded:** Yes
- **Samples per run:** 24 of samples

Computer and Bioinformatics

- **Analysis tools**
- [EPI2ME](#)
- [EPI2ME 16S workflow](#)
- **Software to download or install:**
- [Oxford Nanopore Software Downloads](#) *Oxford Nanopore account and login required:*
- [MinKNOW](#)
- [EPI2ME Desktop Application](#)
- **Other software:**
- [Docker](#)
- **Analysis difficulty:** Easier
- **Command line needed:** No
- **GPU/Super-high accuracy basecalling required:** No

gents**Personal protective equipment**

- As recommended by original protocols (e.g., gloves, lab coat)

Sample collection and prep

- [Norgen Swab Collection and DNA Preservation System](#)
- *Included*
- Sterile swab
- DNA preservation tube

DNA extraction

- [Norgen Microbiome DNA Isolation Kit](#)
- *Included*
- Lysis Additive A
- Binding buffer I
- 70% Ethanol
- Binding Buffer B
- Wash Solution A
- Elution Buffer B
- *User-provided*
- 96-100% ethanol
- 70% ethanol

DNA prep, library creation, and sequencing

- *Nanopore kit*
- *Included*
- 16S Barcodes in 96-well plate, at 1 μ M each
- EDTA
- AMPure XP Beads (AXP)
- Elution Buffer (EB)
- Rapid Adapter (RA)
- Adapter Buffer (ADB)
- *User-provided*
- LongAmp Hot Start Taq 2X Master Mix (NEB, M0533)
- Nuclease-free water
- Freshly prepared 80% ethanol in nuclease-free water
- Qubit dsDNA HS Assay Kit (Invitrogen, Q32851) optional
- *DNA quality control*
- If using Qubit
- Qubit dsDNA HS Assay Kit (Invitrogen, Q32851) optional
- If using electrophoresis
- Electrophoresis grade agarose
- TBE buffer
- GelRed/SYBR Green, or other DNA dye for visualization
- 100bp DNA ladder

Equipment and consumables

Lab equipment

- PCR thermocycler
- Microcentrifuge (20,000 x g)
- Micropipette set (e.g., P10, P100, P1000)
- Assorted tube racks (microfuge and PCR tubes)
- Magnetic rack for 1.6-1.7ml tubes
- Ice bucket with ice
- Electrophoresis chamber, power supply, and documentation setup OR Qubit fluorometer
- (Optional) Hula mixer
- (Optional) Vortexer
- Permanent markers

Consumables

- 1.5-1.7ml microfuge tubes
- 0.2 ml thin-walled PCR tubes
- (Optional) 1.5 ml Eppendorf DNA LoBind tubes

Nanopore sequencing equipment

- Sequencing device: [MinION](#) sequencer (M1kB,C, or D) with [Flongel](#) adapter and [Flongle](#) flow cells

Computer equipment

- Desktop or laptop with [MinKNOW](#) and [EPI2ME](#) installed

Estimated timings

- **Collection of microbes from shoe swab:** 10 minutes
- **DNA extraction:** 60-75 minutes
- **PCR, DNA quality control, and library prep:**
 - **PCR:** 3-3.5 hours
 - **Quality control:** 10 min (Qubit); 60-75 min (electrophoresis)
 - **Library prep:** 30-60 min
- **Sequencing:** 60 min to 24 hours
- **Data analysis:** 30-60 min

Background

 [Playlist: 16S Shoe-ome Experiment](#) 

Sequencing 16S DNA provides insights into the composition and diversity of microbial communities by targeting the 16S ribosomal RNA (rRNA) gene, (1) a conserved region present in all bacteria. This gene contains variable regions that can differentiate bacterial species and genera, making it a powerful tool for identifying and classifying bacteria in a sample. By analyzing 16S DNA sequences, researchers can determine which bacterial taxa are present, estimate their relative abundances, and compare microbial communities across different environments or conditions. This method is widely used in microbiome studies, environmental monitoring, and pathogen identification.

1. Read more about 16S [Wikipedia](#)

In the "Shoe-ome" experiment, students swab the exterior bottom of their shoes—an easily obtained, personalized sample with minimal risk of accidentally collecting sensitive DNA. High school students with limited or no laboratory experience can complete this experiment in a day-long, 8-hour lab session or across two shorter sessions. DNA is collected from a swab sample, lysed, and concentrated using a commercial kit. The 16S region is amplified, barcoded, and sequenced using the Nanopore kit. The entire 16S rRNA gene is amplified via PCR with barcoded primers, allowing the multiplexing of up to 24 samples in a single sequencing run. The EPI2ME workflow analysis generates several visualizations and characterizations of the diversity and types of microbes detected.

Additional Reading

1. Evaluation of 16S rRNA gene sequencing for species and strain-level microbiome analysis: [paper](#)
2. Performing accurate species-level bacterial identification with nanopore sequencing: [paper](#)
3. How Bacteria Rule Over Your Body – The Microbiome: [video](#) 

Collection of microbes from shoe swab

Goal: Obtain a microbial sample for DNA sequencing using the swab kit to sample the bottom of a shoe.(1)

1. Video: [Educator's Guide to Nanopore: Shoe DNA swab](#) 



Instruction tip

Each student (up to 24) can prepare their own sample.

1. Obtain a shoe you wish to sample. The shoe does not need to be visibly dirty; however, a shoe worn daily is likely to have greater microbial diversity than one that has never been worn.
2. Carefully unscrew the cap of the collection tube without spilling the contents. Next, carefully remove the sterile swab from its package, being cautious not to touch the swab or contaminate it in any way. Dip the swab into the liquid in the collection tube.
3. Use the moistened swab to gently rub the underside and sides of the shoe. A moist swab is more effective at collecting microbes that may be present on the shoe. Ideally, the swab will visibly change in appearance as it picks up dirt and microbes.
4. Return the swab to the collection tube. The swab has a "score," a thinned section of the plastic that allows you to break the swab in half, with the half remaining in the tube fitting just enough to be enclosed.
5. Replace the cap on the collection tube, ensuring that the tube is fully closed and leakproof. The solution in the tube deactivates any microorganisms and preserves DNA. Label the tube with your name and date.

DNA extraction

Goal: Extract and purify DNA from the collected microbes to use in PCR using the microbiome DNA extraction kit.(1)

1. Video: [Educator's Guide to Nanopore: DNA extraction](#) 



References

This protocol follows *Protocol C* from the Norgen [Swab Collection and DNA Preservation System](#).

Part I - Preparing swab sample

1. Add 100 μL of Lysis Additive A to the swab collection tube and vortex briefly.

**Sample bias**

The lysis additive and heating steps are meant to break open microbes allowing their DNA to be extracted. Some microbes are tough and may not be efficiently lysed. Keep in mind that difficult-to-lyse or microbes and very rare microbes will be less likely to be detected in your dataset, even if present.

2. Incubate the swab collection tube at 65°C for 5 minutes.
3. Carefully remove the swab from the collection tube.
4. Label a microcentrifuge tube and transfer up to 1 mL of the preserved sample to the microcentrifuge tube.
5. Centrifuge the tube for 2 minutes at $20,000 \times g$ (~14,000 RPM). A thin white layer will form on the top of the supernatant.
6. Label a new microcentrifuge tube and carefully transfer 700 μL of supernatant, without the white layer debris, to the microcentrifuge tube.
7. Add 100 μL of Binding Buffer I, mix by inverting the tube a few times, and incubate for 10 minutes on ice.
8. Spin the lysate for 2 minutes at $20,000 \times g$ (~14,000 RPM) to pellet any cell debris.
9. Label a new microcentrifuge tube. Using a pipette, transfer up to 700 μL of supernatant (avoid contacting the pellet with the pipette tip) into the new microcentrifuge tube.
10. Add an equal volume of 70% ethanol to the lysate collected above (100 μL of ethanol is added to every 100 μL of lysate). Vortex to mix.

Part II - binding DNA to column

1. Assemble a spin column with one of the provided collection tubes.
2. Apply 700 μL of the clarified lysate with ethanol onto the column and centrifuge for 1 minute at $10,000 \times g$ (~10,000 RPM). Discard the flowthrough and reassemble the spin column with the collection tube.

**Tip**

Ensure the entire lysate volume has passed through into the collection tube by inspecting the column. If the entire lysate volume has not passed, spin for an additional minute at $20,000 \times g$ (~14,000 RPM).

3. Repeat step 2 with the remaining volume of lysate mixture.

Part III - wash column

1. Apply 500 μL of Binding Buffer B to the column and centrifuge for 1 minute at $10,000 \times g$ (~10,000 RPM).

**Tip**

Ensure the entire Binding Buffer B has passed through into the collection tube by inspecting the column. If the entire wash volume has not passed, spin for an additional minute.

2. Discard the flowthrough and reassemble the spin column with its collection tube.
3. Apply 500 μL of Wash Solution A to the column and centrifuge for 1 minute at $10,000 \times g$ (~10,000 RPM).
4. Discard the flowthrough and reassemble the spin column with its collection tube.
5. Repeat steps 3 and 4.
6. Spin the column for 2 minutes at $20,000 \times g$ (~14,000 RPM) in order to thoroughly dry the resin. Discard the collection tube.

Part IV - Elute DNA

1. Place the column into a fresh 1.7 mL Elution tube provided with the kit.
2. Add 50 µL of Elution Buffer B to the column.
3. Centrifuge for 1 minute at 425 x g (~2,000 RPM), followed by a 1 minute spin at 20,000 x g (~14,000 RPM). If the entire volume has not been eluted, spin the column at 20,000 x g (~14,000 RPM) for 1 additional minute.

**Optional**

An additional elution may be performed if desired by repeating steps 2 and 3 using 50 µL of Elution Buffer. The total yield can be improved by an additional 20-30% when this second elution is performed.

The purified genomic DNA can be stored at 2-8°C for a few days. For longer term storage, -20°C is recommended.

**Pause point**

Once DNA extraction is completed you can stop here. However, prepping the PCR can be done in under 20 minutes and since the PCR is long, you may wish to move ahead so that the PCR can run overnight or between lab sessions.

PCR, DNA quality control, and library prep

Goal: Amplify the 16S region from the purified shoe DNA sample and prepare it for sequencing on a flow cell.(1)

1. Video: [Educator's Guide to Nanopore: PCR](#) 

**Nanopore**

This protocol follows the **Flongel** version of [Rapid sequencing DNA - 16S Barcoding Kit 24 V14 \(SQK-16S114.24\)](#) from Oxford Nanopore.

**DNA quantification**

The Nanopore protocol requires 10 ng of high molecular weight genomic DNA per barcode. A Qubit fluorometer or Nanodrop spectrophotometer can be used to quantify this small amount of DNA. However, in practice, we often proceed to PCR without quantification, acknowledging the increased risk that some samples may have yields too low to generate a successful PCR product.

**Instruction tip**

Generally, we recommend the lab instructor prepare and pool the individual samples on the student's behalf. We usually narrate the steps and make use of web cams to allow students to have a close up view.

1. Take one 96-well plate containing 16S barcodes. Break one set of barcodes (1-24, or as desired) away from the plate and return the rest to storage.
2. Thaw the desired barcodes at room temperature.
3. Briefly centrifuge barcodes in a microfuge to make sure the liquid is at the bottom of the tubes and place on ice. If you don't have a plate centrifuge, tap the plate several times to settle the liquid to the bottom.
4. Thaw the LongAmp Hot Start Taq 2X Master Mix, spin down briefly, mix well by pipetting and place on ice.
5. For each Shoe DNA sample, add 15µl of sample to a 0.2ml PCR tube. If you have quantified samples, use nuclease-free water to adjust your DNA samples so that your 15µl sample has a total of 10 ng of DNA.



In a classroom situation, we regularly skip quantification. We also will generally try samples that have a lower-than-optimal concentration of DNA. In general, it is better use err on the side of caution using less DNA rather than overloading with more.

6. In each 0.2 ml thin-walled PCR tube containing a sample to be tested, prepare the following mixture:

Reagent	Volume
10 ng input DNA (from previous step)	15µl
LongAmp Hot Start Taq 2X Master Mix	25µl
Total	40µl

7. Ensure the components are thoroughly mixed by pipetting and spin down briefly.
8. Using clean pipette tips, carefully pierce the foil surface of the required barcodes. Use a new tip for each barcode to avoid cross-contamination. Make a note of which barcode numbers will be run for each sample.
9. Mix the 16S barcodes by pipetting up and down 10 times. Transfer 10 µl of each 16S Barcode into respective sample-containing tubes.
10. Ensure the components are thoroughly mixed by pipetting the contents of the tubes 10 times and spin down.



Mix gently to minimize introducing air bubbles to the reactions.



Adding barcodes

Each of the 16S primers in the plate contains one of 24 barcodes. During amplification, PCR products will incorporate a unique barcode sequence and a sequence for the rapid adapter chemistry.

11. Amplify using the following cycling conditions:

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	1 min	1
Denature	95 °C	20 sec	3
Anneal	50 °C	30 sec	3
Extend	65 °C	2 min	3
Denaturation	95 °C	20 sec	38
Annealing	55 °C	30 sec	38
Extension	65 °C	2 min	38
Final Extension	65 °C	5 min	1
Hold	4 °C	Infinite	-



Tip

Mix gently to minimize introducing air bubbles to the reactions.



Note

With LongAmp Taq, this PCR can take 3 hours or more. Once the PCR reaction is started you can stop here and proceed with [library preparation](#) in a subsequent lab session.



Flow cell check

When you are ready to prepare the [library](#) (30-45 min), you should proceed directly to loading the [flow cells](#). Now is a good time to [check the quality of your flow cell](#).



Educator's Guide to Nanopore: [Library preparation](#)



12. Thaw reagents at room temperature, spin down briefly using a microfuge and mix by pipetting as indicated by the table below:

Reagent	Thaw at Room Temperature	Briefly Spin Down	Mix Well by Pipetting or Vortexing
Rapid Adapter (RA)	Not frozen	✓	Pipette
Adapter Buffer (ADB)	✓	✓	Vortex or Pipette
AMPure XP Beads (AXP)	✓	✓	Mix by vortexing immediately before use
Elution Buffer (EB)	✓	✓	Vortex or Pipette
EDTA (EDTA)	✓	✓	Vortex or Pipette

13. Add 4 µl of EDTA to each barcoded sample, mix thoroughly by pipetting and spin down briefly.



Info

EDTA is added at this step to stop the reaction

14. Incubate for 5 minutes at room temperature.

15. Check your PCR product

Qubit quantification Electrophoresis quantification

Follow the [dsDNA High Sensitivity](#) protocol to verify the concentration of the product. While you can obtain an absolute quantification, you may still want to examine products via electrophoresis to verify the size of the fragment and the presence of artifacts (e.g primer dimer, spurious products).

You can run PCR products on a 1.5% agarose gel to verify the product and obtain a relative quantification.

16. Pool all barcoded samples in equimolar ratios in a 1.5 ml tube.



Optional

Eppendorf DNA LoBind tubes are recommended.

17. Resuspend the AMPure XP Beads (AXP) by vortexing.

18. To the pool of barcoded samples, add a 0.6X volume ratio of resuspended AMPure XP Beads (AXP) and mix by pipetting:

Volume of Barcoded Sample Pool	37.5µl	75µl	150µl	300µl	600µl
Volume of AMPure XP Beads (AXP)	22.5µl	45µl	90µl	180µl	360µl

19. Incubate for 5 minutes at room temperature; gently mix by inversion every minute or so; or use a Hula mixer.

20. Prepare 2 ml of fresh 80% ethanol in nuclease-free water.

21. Briefly spin down the sample and pellet on a magnetic rack until supernatant is clear and colorless. Keep the tube on the magnetic rack, and pipette off the supernatant.

22. Keep the tube on the magnet and wash the beads with 1 ml of freshly-prepared 80% ethanol without disturbing the pellet. Remove the ethanol using a pipette and discard.

23. Repeat the previous step.

24. Spin down and place the tube back on the magnet. Pipette off any residual ethanol. Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.

25. Remove the tube from the magnetic rack and resuspend the pellet by pipetting in 15 µl Elution Buffer (EB). Spin down and incubate for 5 minutes at room temperature.

26. Pellet the beads on a magnet until the eluate is clear and colorless, for at least 1 minute.

27. Remove and retain 15 µl of eluate into a clean 1.5 ml tube (Eppendorf LoBind if using).



Optional

Nanopore recommends here you quantify 1 µl of eluted sample using a Qubit fluorometer. In our experience you may omit this step if you don't have a Qubit.

If you do calculate the concentration, you can meet the loading concentration recommendations (often in femtomoles) using a [calculator](#) and assuming a ~1,500bp amplicon.

28. In a fresh 1.5 ml tube (Eppendorf LoBind if using), dilute the [Rapid Adapter \(RA\)](#) as follows and pipette mix:

Reagent	Volume
Rapid Adapter (RA)	1.5µl
Adapter Buffer (ADB)	3.5µl
Total	5µl

29. Add 0.5µl of diluted [Rapid Adapter \(RA\)](#) to the barcoded DNA.

30. Mix gently by flicking the tube, and spin down.

31. Incubate the reaction for 5 minutes at room temperature. Then place the final product on ice while you prepare for loading the [flow cell](#).

**Proceed directly to loading**

We recommend you proceed to loading your [flow cell](#) and sequencing, since the prepared [library](#) works best, in our experience, if it is used shortly after prep.

Sequencing

Goal: Load prepared [library](#) onto [flow cell](#) and sequence.(1)

1. Video: [Educator's Guide to Nanopore: Flongle flow cell loading](#) 



This protocol follows the **Flongel** version of [Rapid sequencing DNA - 16S Barcoding Kit 24 V14 \(SQK-16S114.24\)](#) from Oxford Nanopore.

1. Thaw the Sequencing Buffer (SB), [Library Beads \(LIB\)](#) or [Library Solution \(LIS\)](#), if using, Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.
2. In a fresh 1.5 ml tube (Eppendorf LoBind if using), mix 117 µl of Flow Cell Flush (FCF) with 3 µl of Flow Cell Tether (FCT) and mix by pipetting.
3. Place the [Flongle](#) adapter into the [MinION](#).
4. Place the [flow cell](#) into the [Flongle](#) adapter, and press the [flow cell](#) down until you hear a click.
5. Peel back the seal tab from the [Flongle flow cell](#), up to a point where the sample port is exposed.
6. To prime your [flow cell](#) with the mix of Flow Cell Flush (FCF) and Flow Cell Tether (FCT) that was prepared earlier, ensure that there is no air gap in the sample port or the pipette tip. Place the P200 pipette tip inside the sample port and slowly dispense the 120 µl of priming fluid into the [Flongle flow cell](#) by slowly pipetting down. We also recommend twisting the pipette plunger down to avoid flushing the [flow cell](#) too vigorously.

**Tip**

The [Library Beads \(LIB\)](#) tube contains a suspension of beads. These beads settle very quickly. It is vital that they are mixed immediately before use.

7. Vortex the vial of [Library Beads \(LIB\)](#). Note that the beads settle quickly, so immediately prepare the Sequencing Mix in a fresh 1.5 ml Eppendorf DNA LoBind tube for loading the [Flongle](#), as follows:

Reagents	Volume
Sequencing Buffer (SB)	15 µl
Library Beads (LIB) (mixed immediately before use) or Library Solution (LIS) (if using)	10 µl
DNA Library	5 µl
Total	30 µl

8. To add the Sequencing Mix to the [flow cell](#), ensure that there is no air gap in the sample port or the pipette tip. Place the P200 tip inside the sample port and slowly dispense the Sequencing Mix into the [flow cell](#) by slowly pipetting down. We also recommend twisting the pipette plunger down to avoid flushing the [flow cell](#) too vigorously.
9. Seal the [Flongle flow cell](#) using the adhesive on the seal tab. Close the [MinION](#) and proceed to start the sequencing run.

**Nanopore**

This protocol follows the **Flongle** version of [Rapid sequencing DNA - 16S Barcoding Kit 24 V14 \(SQK-16S114.24\)](#) from Oxford Nanopore.

**Informatics**

For the following steps, you will use the [MinKNOW software](#) to operate the [MinION](#) device.

10. Navigate to the start page and click Start sequencing.
11. Fill in your experiment details, such as name and [flow cell](#) position and sample ID.
12. Select the sequencing kit (16S Barcoding Kit 24 V14 (SQK-16S114.24)) used in the [library preparation](#) on the Kit page.

13. Configure the sequencing parameters for your sequencing run or keep to the default settings on the Run options and Analysis tabs.

Computer with GPU

Computer without GPU

We recommend:

- Raw reads: [.POD5](#)
- Basedcalled reads: [FASTQ](#)
- Basecalling: High-accuracy basecalling (HAC) or Super-accurate basecalling (SUP)
- Modified bases: Off

We recommend:

- Raw reads: [.POD5](#)
- Basedcalled reads: [FASTQ](#)
- Basecalling: Fast [basecalling](#)
- Modified bases: Off



Tip

You may also leave all [basecalling](#) off if you will perform that operation on a cloud platform or more powerful computer.

14. On the Output page, set up the output parameters or keep to the default settings.

15. Click Start on the Review page to start the sequencing run.



Note point

In most cases, you can generate useful data to analyze within an hour. A more complete dataset will take about 24 hours (the usual lifetime of a [Flongle](#)).

Data analysis

Goal: Determine the microbes present in the shoe sample and their taxonomic relationships.



Example data

Guam Shoe-ome:

This dataset was generated at a workshop at [JFK High School](#), March 2024.

- Nanopore sequencing report example: [Download](#)
- FastQ files: [Download](#)
- Example [EPI2ME](#) report: [Download](#)



Informatics

This tutorial will use the [EPI2ME 16s workflow](#) developed by Oxford Nanopore.

Workflow version: 1.1.3

**Instruction tip**

Students can work individually or in groups to analyze their data. Assuming each student sample is assigned their own barcoded samples, they can generate an individual report specific to their example.

ANALYZE DATA

1. Open the [EPI2ME](#) software and choose either **Sign in** or **Continue as guest**.

**Informatics**

In this example, we use the **Continue as guest** option.

2. Click on the **Launch** icon and choose the **16S** workflow.

**Informatics**

In this example, choose **Run Locally**.

**Failure**

If you see the "Setup required" error message, follow the instructions to install or configure any needed software (for example, installing and starting [Docker](#)).

**Option**

Depending on your access and your computer, the launch page is where you will decide on running the workflow locally (on your computer) or on the cloud.

3. Click on the **Launch** button to proceed with the workflow.
4. Under **Setup local analysis**, complete the required options to load your sequencing dataset and choose analysis parameters.

**Informatics**

In this example, we use the following parameters:

Input Options

- *FASTQ*: "fastq_pass" folder provided in this tutorial example data. This folder will be generated by your experiment.
- *Bam*: Leave as default (blank).
- *Classification method*: Leave as default (minimap2).
- *Analyze unclassified reads*: Leave as default (unchecked).
- *Exclude host reads*: Leave as default (blank).

Real-Time Analysis Options

- All options left as default.

**Tip**

You can start your EPI2ME analysis while sequencing is ongoing. This allows you to show a real-time analysis of your data rather than waiting until sequencing is completed.

Sample Options

- All options left as default.

Reference Options

- *Choose a database*: "SILVA_138_1".
- All other options left as default.

Kraken2 Options

- All options left as default.

Minimap2 Options

- *Compute coverage and sequencing depth of the references*: Checked.
- All options left as default.

Report Options

- All options left as default.

Advanced Options

- All options left as default.

Miscellaneous Options

- All options left as default.

Nextflow Options

- All options left as default.

5. Click **Launch workflow** and then **Launch** again.

**Time**

The time required for this analysis depends on your dataset. On a computer with the recommended configuration for running a MinION, the test dataset was completed in approximately 35 minutes.

INTERPRET RESULTS

The wf-16s pipeline generates a detailed report ([wf-16s-report.html](#)) that provides comprehensive insights into sequencing data and microbial community composition. Below are the major components of the report:

1. **Read Statistics Summary**

This section presents an overview of the sequencing reads, including:

- Total read counts
- Quality metrics
- Length distributions

These metrics help assess the overall quality of the sequencing run.

2. **Taxonomic Composition**

Displays the identified taxa within the sample, often visualized through:

- **Bar plots** or **pie charts** illustrating the relative abundances of different microbial groups.

3. **Diversity Metrics**

Includes calculations of:

- **Alpha diversity** indices, such as [Shannon](#) and [Simpson indices](#), which assess diversity within a sample.
- **Beta diversity** analysis for comparing differences between multiple samples (if applicable).

4. **Reference Alignment Statistics**

For workflows using [Minimap2](#) for taxonomic classification, this section provides:

- [Alignment](#) statistics, including sequencing depth and coverage for each reference genome.
- Heatmaps visualizing sequencing depth across genomic coordinates for the most covered references.

5. **Rarefaction Curves**

Plots display the mean species richness as a function of sampling effort, helping to determine if the sequencing depth was sufficient to capture the sample's diversity.

Learn about [rarefaction curves](#).

6. **Abundance Tables**

Provides detailed tables listing the counts of different taxa per sample.

- Customizable filters (e.g., *abundance_threshold*) can exclude low-abundance taxa for clearer analysis.

These components collectively offer a robust analysis of sequencing data, enabling a deeper understanding of microbial communities and their diversity.



Instruction tip

In depth explanations of the report metrics are available on the [16S workflow github documentation page](#).



THINGS TO LOOK FOR

There are a variety of results contained in the 16S report. While it is not possible to describe all outcomes there are several things to draw your attention to.

• Read Quality

- Uncorrected Nanopore reads have a lower quality score, especially with fast basecalling. You can expect uncorrected reads in the 8-12 phred score range. Using super accurate basecalling is one way you can improve this result.

• Read Length

- A sharp peak around 1500bp—the length of the 16S amplicon—should be expected.

• Samples summary

- Ideally, you will have an about equal number of reads from all barcodes. This is difficult to achieve, and barcodes with lower reads will be under-sampled relative to more abundant samples. You could prepare a new library to adjust if something critical was underrepresented.
- You will occasionally get a few (<10) reads from barcodes that were not used in your experiment. Sequencing error may misidentify a barcode, misclassifying it. This is normal.

Please enable JavaScript to view the [comments powered by Disqus](#).

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🕒 January 6, 2025

5.3 Curated Experiment: Microbial Whole Genome Sequencing

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5.4 Curated Experiment: Small Plant Genome Sequencing

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5.5 Environmental DNA

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5.6 Phage Genome Sequencing

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6. Bioinformatics

6.1 Data Organization and Management in Nanopore Sequencing

6.1.1 The importance of good data management

Effective data management is essential for ensuring the integrity, reproducibility, and long-term accessibility of sequencing data. Following best practices for organizing, storing, and documenting sequencing data benefits both research efficiency and educational experiences. In genomics, the volume of data generated can be substantial, making structured data organization critical.

A guiding framework for good data stewardship is the **FAIR Principles**:

- **Findable** – Data should be easily discoverable using standardized metadata.
- **Accessible** – Data should be retrievable via established repositories and protocols.
- **Interoperable** – Data should be compatible with various analysis tools and software.
- **Reusable** – Data should be well-documented and structured to allow for future use.

For educators, introducing students to data hygiene—including proper file naming, directory structuring, and metadata collection—helps instill best practices that are critical for careers in bioinformatics, computational biology, and genomics.



Instruction tip

Managing Student-Generated Data

Managing student-generated data in a Nanopore sequencing course requires careful planning to ensure smooth data sharing, organization, and long-term usability. Educators should establish a clear data-sharing strategy that facilitates seamless file exchange between instructors and students, particularly given the large file sizes and/or large number of files associated with sequencing data. Cloud storage solutions like [CyVerse](#), [Google Drive](#), or institutional servers may be useful for storing and distributing datasets, but file size limitations and access controls should be considered. Additionally, educators should anticipate the scientific value of student-generated data—insisting on good data practices from the beginning can ensure that sequencing results are valid, reproducible, and valuable in aggregate.

Developing a **classroom data management plan** can help structure how data are named, stored, and tracked across multiple course sections, lab groups, or project days. Establishing standardized file-naming conventions that incorporate identifiers such as course section, date, group number, and experiment type will allow educators to efficiently organize and retrieve data. Collecting consistent metadata across student projects is also essential if data are to be meaningfully analyzed or aggregated for further research or public sharing. Resources such as the [FAIR Principles](#) provide guidance on making data findable, accessible, interoperable, and reusable.

File and object names and metadata to keep track of from the start

- **Project name:** At the most encompassing level of organization, have a simple project name
- **Students involved:** Have a way of tracking students by more than give names and initials. Perhaps all students can be assigned a short unique identifier. You could consider having students create and use [ORCIDs](#).
- **Sample names:** Have a way to name specific samples. The sample name might also include or be linked to acquisition information such as the date collected, or the location of sampling. If a single sample is sampled multiple times, have a way to track and indicate this.
- **Replicates:** Have a way to indicate replication.
- **Data directory structures:** Consider how directories on storage platforms will be organized. Have rules for how folders should be named, when folders should be created, and what should be in them.
- **Collision avoidance:** Some files and folders will not have unique names. For example, every time you run a 16S workflow on [MinKNOW](#), you will get folders named `barcode01`, `barcode02`, `barcode03`... Know when you will encounter these and have a way to manage their organization, or if needed, renaming, to avoid confusion.
- **Software management:** Have a way to keep track of the software versions of any software you will use.

While this list is not exhaustive, considering these possibilities ahead of time will make for smooth organization later.

By integrating good data hygiene and documentation practices, educators can help students engage with sequencing workflows in a way that mirrors professional research environments.

Data management plans

A **Data Management Plan (DMP)** is an essential document that outlines how data will be:

- Collected
- Stored and backed up
- Processed and analyzed
- Shared or archived

Many funding agencies require DMPs to ensure that research data remains well-documented and reusable. While formal DMPs may not be necessary for classroom projects, educators should emphasize the importance of:

- Keeping consistent directory structures
- Using clear, standardized file naming conventions
- Documenting key metadata
- Implementing proper backup procedures



Additional Reading

1. **National Human Genome Research Institute.** (n.d.). [Genomic Data Science](#). National Human Genome Research Institute (NHGRI).
2. **Oza, V. H., Whitlock, J. H., Wilk, E. J., Uno-Antonison, A., Wilk, B., Gajapathy, M., Howton, T. C., Trull, A., Ianov, L., Worthey, E. A., & Lasseigne, B. N.** (2023). [Ten simple rules for using public biological data for your research](#). *PLoS Computational Biology*, 19(1), e1010749. <https://doi.org/10.1371/journal.pcbi.1010749>.
3. **Form, D., & Lewitter, F.** (2011). [Ten Simple Rules for Teaching Bioinformatics at the High School Level](#). *PLoS Computational Biology*, 7(10), e1002243. <https://doi.org/10.1371/journal.pcbi.1002243>.
4. **Noble, W. S.** (2009). [A Quick Guide to Organizing Computational Biology Projects](#). *PLoS Computational Biology*, 5(7), e1000424. <https://doi.org/10.1371/journal.pcbi.1000424>.

6.1.2 Common file types in Nanopore sequencing

Nanopore sequencing generates multiple types of data files at different stages of the workflow. Below is an overview of the key file formats:

Raw data

- **POD5 (.pod5)** [\[Documentation\]](#)
Current standard format for storing raw signal data from Nanopore sequencing. Replaces the older [FAST5](#) format for improved performance and scalability.
- **FAST5 (.fast5)** [\[Legacy\]](#)
Used in older Nanopore sequencing workflows to store raw signal data. This format is no longer in use, having been replaced by [POD5](#).

Processed data

- **FASTQ (.fastq)** [\[Documentation\]](#)
Primary file format for storing basecalled sequence reads, including nucleotide sequences and associated quality scores.
- **BAM (.bam) / CRAM (.cram)** [\[Documentation\]](#)
Binary formats used to store sequence alignments to a reference genome. CRAM is a more compressed version of BAM.

Reference data

- **FASTA (.fasta / .fa)** [\[Documentation\]](#)
Stores nucleotide or protein sequences without quality scores. Often used for reference genomes or consensus sequences.
- **GFF/GTF (.gff / .gtf)** [\[Documentation\]](#)
Contains genome annotations, such as gene locations and functional elements.

Analysis and metadata

- **CSV/TSV (.csv / .tsv)**
Tabular files commonly used for summarizing sequencing statistics, barcode classifications, or other metadata.
- **JSON (.json)**
Structured format for storing run configurations, analysis parameters, or experiment details.
- **VCF (.vcf)**
Variant Call Format, used for storing genetic variations detected in sequencing data.

Reports

- **HTML (.html)**
Many analysis pipelines generate interactive reports in HTML format, offering visual summaries, charts, and metrics. This format is useful for sharing final results with collaborators or for archiving purposes.

6.1.3 Managing genomics data for long-term use

Metadata collection

Metadata provides essential **context** for sequencing datasets and is often required for submitting data to public repositories like the [NCBI Sequence Read Archive \(SRA\)](#). Key metadata elements may include:

- **Sample ID** (Unique identifier for the sample)
- **Collection date** (When the sample was obtained)
- **Location** (Geographic origin of the sample)
- **Extraction and library prep methods** (Protocols used)
- **Sequencing platform and flow cell type**
- **Bioinformatics processing steps** (Pipeline details, tools used)

Storage and backup strategies

- **Primary Storage:** A local high-capacity SSD (recommended 1TB or more) for sequencing and short-term analysis.
- **Backup Storage:** Redundant copies using external SSDs or network drives.
- **Cloud Solutions:** Platforms like Google Drive, CyVerse, or JetStream2 for scalable storage.
- **Public Repositories:** Depositing finalized datasets in SRA, EMBL-EBI ENA, or MG-RAST for community access.

Preparing data for public submission

- Ensure [FASTQ](#) files are properly formatted.
- Validate metadata against repository requirements.
- Use consistent file naming conventions (e.g., `sampleID_experimentType_date.fastq`).
- Include a README file describing dataset contents and processing steps.

6.1.4 Quickstart: How to find and download Nanopore reads from the Sequence Read Archive (SRA)

The [NCBI Sequence Read Archive \(SRA\)](#) is a public repository that stores sequencing data, including Oxford Nanopore sequencing reads. This tutorial provides a step-by-step guide to searching for data, in this case, **Actinomyces** sequences generated using **Oxford Nanopore** technology and downloading a [FASTQ](#) file for

further analysis. You can adapt the tutorial to search for data from your organism of interest and use those data for reanalysis and to practice your bioinformatics skills.

1. Open your web browser and visit the [NCBI SRA website](#).
2. Click on the **Advanced** link beneath the search bar to access the detailed search options.

[SRA homepage](#)

3. In the first box of the search **Builder**, type **Actinomyces** and click the **Show index list** link. The number in parentheses shows how many records match the term. You can click on the term of choice, in this case **actinomyces (936)**, and select **Organism** from the dropdown to ensure that is the level being searched.

[SRA advanced search - Organism](#)

4. In the second box of the search **Builder**, type **Oxford Nanopore** and click the **Show index list** link. The number in parentheses shows how many records match the term. You can click on the term of choice, in this case **oxford nanopore (875940)**, and select **Platform** from the dropdown to ensure that is the level being searched.

[SRA advanced search - Platform](#)

5. Click the **Search** button to retrieve relevant sequencing datasets.
6. Browse through the search results and look for a **16S sequencing** dataset. In this case, we choose a selection marked **16S rRNA sequence of isolates**. Note the number of bases (M - megabases, G - gigabases) and size of the dataset in Mb/Gb is shown, alerting you to the size of the potential download.; Click on a result to view the dataset details.

[SRA search results](#)

7. The results page will have various metadata including an abstract for the entry which may contain useful background on the experiment that produced the data.



Tip

The **All experiments** or **All runs** link will help you find connected datasets which may have come from other libraries or samples in this same experiment.

8. Locate the **Runs** table, which lists individual sequencing runs associated with the dataset.
9. Click on the **Run Accession** link (e.g., **SRRXXXXXXX**) of a single sequencing run to open its details page.

[Accession link](#)

10. On the run browser page, click the **FASTA/FASTQ download** link.

[Download link](#)

11. Under **Download** click on **FASTQ** to start a download for your file. The file will be in a compressed **.fastq.gz** format. This is a typical input for many downstream analyses.

[Fastq link](#)

Additional Notes

- The downloaded FASTQ file contains raw sequence reads and quality scores that can be used for taxonomic classification, assembly, or other bioinformatics workflows.

- Advanced users may explore **SRA Toolkit**, a command-line software package that enables more efficient searching, downloading, and processing of SRA data. The toolkit provides options such as `prefetch` for downloading and `fastq-dump` for extracting sequence reads. More details can be found in the [SRA Toolkit documentation](#).

🕒 February 10, 2025

🕒 February 4, 2025

6.2 Software Installation and Management

6.2.1 Installing essential software for Nanopore sequencing

To operate a Nanopore sequencer and analyze sequencing data, three key software tools are required:

- **MinKNOW**: The primary control software for Oxford Nanopore sequencing devices.
- **EPI2ME**: A cloud-based or local analysis platform for processing sequencing data.
- **Docker**: A containerization platform required for running some local bioinformatics workflows.

Below are step-by-step instructions for downloading and installing each tool on **macOS** and **Windows**.

6.2.2 MinKNOW

MinKNOW is used to control Nanopore sequencing devices, perform **basecalling**, and monitor sequencing progress.

macOS Installation	Windows Installation
--------------------	----------------------

- | | |
|---|---|
| <ol style="list-style-type: none">1. Visit the Oxford Nanopore Downloads Page and choose your device.2. Select MacOS for Apple or Intel (open About This Mac, choose Apple menu  > About This Mac to check the Chip listed.); click Download.3. Sign in to your Oxford Nanopore account (or create one if you don't have an account); your download should begin.4. Open the downloaded <code>.dmg</code> file and drag the MinKNOW application to the Applications folder.5. Open MinKNOW from the Applications folder.6. Follow the on-screen setup instructions and allow necessary permissions when prompted.7. Restart your computer to ensure all components load correctly. | <ol style="list-style-type: none">1. Visit the Oxford Nanopore Downloads Page and choose your device.2. Sign in to your Oxford Nanopore account (or create one if you don't have an account); your download should begin.3. Download the latest MinKNOW for Windows.4. Open the downloaded <code>.exe</code> file and follow the on-screen installation instructions.5. Restart your computer if prompted.6. Open MinKNOW and ensure it launches properly. |
|---|---|
-

6.2.3 Installing EPI2ME

EPI2ME is a workflow management tool used for analyzing Nanopore sequencing data.

macOS Installation Windows Installation

1. Visit the [EPI2ME Labs download page](#).
2. Select macOS for M1/M2 (Apple) or Intel (open About This Mac, choose Apple menu  > About This Mac to check the **Chip** listed.); click **Download**.
3. Open the `.dmg` file and drag the **EPI2ME** application to the Applications folder.
4. Open **EPI2ME** from the Applications folder.
5. Sign in with an existing account or create a new account if needed.
6. Follow the initial setup process to ensure **EPI2ME** is ready for use.

1. Visit the [EPI2ME Labs download page](#).
2. Download **EPI2ME for Windows**.
3. Open the downloaded `.exe` file and follow the on-screen installation instructions.
4. Launch **EPI2ME** and sign in with an existing account or create a new one.
5. Complete the initial setup and confirm that the software runs correctly.

6.2.4 Installing Docker

Docker is required to run many **EPI2ME** workflows locally.

macOS Installation Windows Installation

1. Visit the [Docker website](#) and scroll to find the **Download Docker for Desktop** button. Select Download for Mac MacOS for Apple Silicon or Intel Chip (open About This Mac, choose Apple menu  > About This Mac to check the **Chip** listed.).
2. Open the `.dmg` file and drag the **Docker** application to the Applications folder.
3. Open **Docker** from the Applications folder.
4. Follow the setup process and **allow necessary permissions** when prompted.
5. Once Docker is running, verify the installation by opening a terminal and running:

```
docker --version
```

1. Download **Docker Desktop for Windows**.
2. Open the downloaded `.exe` file and follow the setup instructions.
3. During installation, **enable WSL 2 (Windows Subsystem for Linux)** when prompted. If WSL 2 is not installed, follow the instructions provided by Docker to install it.
4. Restart your computer when the installation is complete.
5. Open **Docker Desktop** from the Start menu.
6. Verify the installation by opening **Command Prompt** and running:

```
docker --version
```

6.2.5 Other software packages for Nanopore sequencing and genomics analysis

Below is a curated list of 40 software tools that are widely used for Nanopore sequencing and genomics data analysis. The selection is based on recent updates in their documentation (within the past two years) and current usage in the field. Each entry includes a brief description and a link to the tool's documentation or repository. Some of these are more advanced in their usage (i.e., commandline) but you are likely to see them or want to learn more about their capabilities.

Below is a curated list of 40 software tools widely used for Nanopore sequencing and genomics data analysis. The tools are categorized based on their primary function to facilitate easy navigation.

6.2.6 Quality Control and Preprocessing

- **FastQC**
Performs quality control checks on raw sequencing data in [FASTQ](#) format.
- **MultiQC**
Aggregates reports from multiple QC tools into a single summary.
- **NanoPlot**
Generates visualizations and statistics for assessing Nanopore sequencing data quality.
- **Seqtk**
A lightweight tool for processing [FASTA/FASTQ](#) files, including format conversion and subsampling.

6.2.7 Basecalling and Signal Processing

- **Dorado**
High-performance basecaller for Nanopore sequencing data.
- **Bonito**
A deep learning-based basecaller for improved accuracy.
- **Nanopolish**
Analyzes raw signal data to refine consensus sequences and detect base modifications.

6.2.8 Genome Assembly

- **Flye**
A de novo assembler optimized for Nanopore and PacBio [long reads](#). Probably the best place to start for assembly with Nanopore reads.
- **Canu**
Assembles genomes using long-read sequencing data.
- **SPAdes**
A versatile genome assembler that supports hybrid assemblies combining short and [long reads](#).
- **Unicycler**
A hybrid assembly pipeline combining short and [long reads](#) for high-quality microbial genome assemblies.

6.2.9 Genome Alignment

- **Minimap2**
An efficient aligner for mapping long sequencing reads to a reference genome.
- **BWA-MEM2**
A faster, more efficient version of BWA-MEM for read alignment.
- **STAR**
A fast aligner optimized for RNA sequencing.

6.2.10 Variant Calling and Genome Polishing

- **Medaka**
Polishes consensus sequences generated from Nanopore sequencing data.
- **Longshot**
A variant caller optimized for long-read sequencing.

- **FreeBayes**
A haplotype-based variant caller for detecting genetic variations.
- **DeepVariant**
Uses deep learning to improve variant calling accuracy.
- **Pilon**
Improves genome assemblies by correcting errors with short-read data.
- **Racon**
A consensus module for polishing genome assemblies from long-read sequencing.

6.2.11 Taxonomic and Metagenomic Analysis

- **Kraken2**
A taxonomic classifier that assigns microbial reads to taxa using k-mer-based algorithms.
- **MEGAN**
Enables taxonomic and functional analysis of metagenomic data.
- **MetaPhlAn**
A tool for profiling microbial communities from metagenomic data.

6.2.12 Genome Visualization and Annotation

- **IGV (Integrative Genomics Viewer)**
A high-performance visualization tool for large-scale genomic data.
- **UCSC Genome Browser**
A web-based genome browser for accessing and visualizing genomic annotations.
- **BEDTools**
A toolkit for genome arithmetic and feature comparisons.
- **SnpEff**
Annotates genetic variants and predicts their functional impact.
- **VCFtools**
A collection of utilities for processing and analyzing Variant Call Format (VCF) files.
- **PLINK**
A toolset for population-based genetic association studies.
- **QUAST**
Evaluates and compares genome assemblies.

6.2.13 Workflow Management and Reproducibility

- **Nextflow**
A workflow management system for scalable and reproducible bioinformatics pipelines.
- **Snakemake**
A workflow engine for creating reproducible bioinformatics pipelines.
- **Bioconda**
A Conda channel that simplifies installation of bioinformatics tools.
- **Galaxy**
A web-based platform for running bioinformatics workflows.
- **CyVerse**
A cloud-based data management and computation platform for large-scale genomics projects.

6.2.14 General-Purpose Computational Tools

- **Docker**
A containerization platform for reproducible bioinformatics environments.
- **Jupyter Notebook**
An interactive computing environment for running and sharing bioinformatics code.
- **RStudio**
A popular IDE for R, widely used in bioinformatics and data visualization.
- **NCBI SRA Toolkit**
A command-line toolset for accessing sequencing data from the NCBI Sequence Read Archive.
- **JetStream2**
A cloud-based computing resource providing access to GPUs and bioinformatics tools.

For further reading see [Awesome Bioinformatics](#).

6.2.15 Building more advanced bioinformatics skills – The Carpentries

The [Carpentries](#) is a global community dedicated to teaching foundational coding and data science skills through hands-on, interactive workshops. Their lessons are particularly valuable for researchers and educators looking to build computational skills for bioinformatics and genomics. The [Software Carpentry](#) and [Data Carpentry](#) initiatives offer structured lessons on command-line tools, version control, and programming languages like Python and R. Lessons are self-paced, and in-person workshops are available on every continent!



Additional Reading

The [Genomics Data Carpentry lessons](#) provide an excellent introduction to managing and analyzing sequencing data, covering topics such as the UNIX command line, working with FASTQ files, quality control, and genome assembly. These lessons are designed for beginners and offer an accessible entry point for educators and students looking to incorporate bioinformatics into their work.

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6.3 Cloud Computing and other Platforms

6.3.1 DNA Subway: A User-Friendly Bioinformatics Platform

[DNA Subway](#) is an intuitive, web-based bioinformatics tool designed to make genomic data analysis accessible to students and educators. Developed by CyVerse, DNA Subway simplifies complex bioinformatics workflows, guiding users through DNA barcoding, genome annotation, and comparative genomics analyses. The platform allows users to analyze sequencing data without requiring advanced computational skills, making it ideal for classroom and research use.

Transition to DNA Subway 2.0

The current version of DNA Subway (**DNA Subway 1.0**) is **sunsetting in June 2025**, and will be replaced with **DNA Subway 2.0**, which is undergoing a complete redesign to better support Nanopore sequencing data analysis. This next-generation platform will include:

- **Expanded DNA barcoding:** Designed to handle data from Nanopore sequencing enabling educators and students to analyze their own barcode sequences.
- **Small genome analysis:** Support for assembling and annotating microbial and other small genomes using Nanopore sequencing data.
- **Enhanced user experience:** A more modern and interactive interface with improved workflow tracking and data visualization.
- **Cloud-Based computing:** Streamlined data processing through JetStream2 computational **, reducing the need for local computing resources.

Learn More

For updates on the transition to DNA Subway 2.0 and new features, visit the [DNA Subway Sunset Page](#). Educators and researchers interested in integrating DNA Subway into their curriculum can also explore [DNA Subway Learning Resources](#) for guides and learning materials.

Check back soon for tutorials!

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6.3.2 Jetstream2: Advanced Cloud Computing for Nanopore Sequencing

Jetstream2 is a cloud-based computing platform designed to provide researchers and educators with flexible, on-demand access to high-performance computing resources. It offers a user-friendly environment that simplifies data analysis, accelerates discovery, and enhances the availability of artificial intelligence (AI) tools across various scientific domains.



Bioinformatics

With training, anything is possible. That said, JetStream2 cloud resources require a fair bit of bioinformatics skill. Make use of their support and the [Nanopore Network](#) Slack community for help.



Introduction to Jetstream 2

Key Features of Jetstream2

- **Scalability:** Jetstream2 delivers 8 petaFLOPS of virtual supercomputing power, allowing users to scale their computational resources according to project needs.
- **User-Friendly Interface:** The platform provides an intuitive interface, enabling users to create virtual machines (VMs) that replicate your local computing environments.
- **Diverse Resource Allocation:** Jetstream2 comprises four distinct resources:
- **Jetstream2 (CPU only):** Standard computational tasks.
- **Jetstream2 Large Memory:** Memory-intensive applications.
- **Jetstream2 GPU:** Tasks requiring graphical processing units.
- **Jetstream2 Storage:** Dedicated storage solutions.

Each resource is allocated individually, allowing users to tailor their environment to specific project requirements.

Advantages for Nanopore Sequencing

Nanopore sequencing generates substantial volumes of data and often requires intensive computational analysis. Jetstream2's infrastructure is particularly well-suited to meet these demands:

- **High-Performance Cloud Computing:** The platform's substantial computational power accelerates data processing tasks such as [basecalling](#), genome assembly, and variant analysis.
- **GPU Access:** Jetstream2 offers access to [GPU](#) resources, which are essential for running advanced machine learning algorithms and accelerating specific bioinformatics applications. To utilize [GPU](#) instances, users must have explicit access to the Jetstream2-[GPU](#) resource.
- **Scalable Storage Solutions:** With a default allocation of 1 TB of storage—expandable upon request—Jetstream2 efficiently manages the large datasets typical of Nanopore sequencing projects.

Getting Started with Jetstream2

To begin using Jetstream2, you will need an allocation (free) from NSF [ACCESS-CI](#):

Utilize the [Getting Started Guide](#) for step-by-step instructions on accessing resources and creating your first VM instance.

For educators, ACCESS-CI offers educational allocations to support teaching initiatives. Detailed information on obtaining an allocation is available on the [ACCESS-CI website](#). Once granted, these allocation credits can be applied to Jetstream2, providing access to its comprehensive computational resources.

By leveraging Jetstream2, researchers and educators can efficiently manage and analyze large-scale Nanopore sequencing data, facilitating advanced bioinformatics research and education.

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6.3.3 Amazon Web Services: A Comprehensive Cloud Platform for Genomics

Amazon Web Services ([AWS](#)) is a leading cloud computing platform offering a vast array of services that provide flexible, scalable, and reliable computing solutions. For advanced users in genomics, AWS presents powerful tools to manage, process, and analyze large-scale sequencing data. However, **AWS is best suited for highly experienced users** who are comfortable with cloud computing concepts, resource management, and cost monitoring.

Key AWS Services for Genomics

- **Amazon Elastic Compute Cloud (EC2):** EC2 provides resizable virtual machines, allowing users to run applications on a secure and scalable infrastructure. In genomics, EC2 instances can be configured with high-performance computing capabilities, including access to **Graphics Processing Units (GPUs)**, which are essential for computationally intensive tasks such as genome assembly, [basecalling](#), and variant analysis.
- **Amazon Simple Storage Service (S3):** S3 offers scalable object storage designed for large volumes of data. Genomics researchers can utilize **S3 buckets** to store raw sequencing data, intermediate files, and final analysis results. The service ensures high durability and availability, making it suitable for the extensive datasets generated by sequencing projects.

Considerations and Getting Started

While AWS provides robust infrastructure, it is essential to be mindful of associated costs. Services like EC2 and S3 operate on a **pay-as-you-go** model, meaning expenses can accumulate rapidly depending on usage. Proper cost management tools, such as AWS Budgets and AWS Cost Explorer, should be used to avoid unexpected charges.

For those new to AWS, the following resources are available:

- **AWS Getting Started Guide:** This guide provides tutorials and step-by-step instructions to help users familiarize themselves with AWS services and best practices.
- **AWS HealthOmics:** A purpose-built service designed to help researchers store, query, and analyze genomic, transcriptomic, and other omics data efficiently.
- **Introduction to AWS for Bioinformatics:** A practical guide for bioinformatics researchers looking to leverage AWS services for high-performance computing and data storage.

Important Warning

AWS is **not recommended for beginners** and should only be used by individuals with **advanced experience in cloud computing and resource management**. Improper use can result in **unexpected high costs**. Those seeking free or low-cost alternatives for cloud computing should consider platforms like [JetStream2](#), which provides cloud-based computational resources for scientific research at no cost to educators and students.

By leveraging AWS's extensive suite of services, experienced users in genomics can build scalable and efficient workflows to handle the complexities of sequencing data analysis.

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6.3.4 Using Jupyter for Interactive Nanopore Analysis

[Jupyter](#) is an open-source tool designed for interactive computing and reproducible research. It provides a flexible environment, including for analyzing Nanopore data, combining code, text, and visualizations in a single document. Educators and researchers can use **Jupyter Notebooks** or **JupyterLab** to conduct bioinformatics workflows, explore sequencing datasets, and develop tutorials that enhance student learning.

Benefits Jupyter

Jupyter Notebooks are widely used in computational biology because they allow users to create, share, and execute self-contained analytical workflows. Jupyter provides:

- **An interactive environment** for processing sequencing data step by step.
- **Documentation alongside analysis**, ensuring clarity in computational workflows.
- **Support for multiple programming languages**, including Python, R, and Bash.
- **Integration with data visualization tools**, allowing real-time plotting of sequencing results.
- **Improved collaboration and reproducibility**, making it easier for students and researchers to share results and rerun analyses.

Installing Jupyter Notebook and JupyterLab

There are two main ways to install Jupyter: Jupyter Notebook and JupyterLab. Jupyter Notebook provides a simplified interface, while JupyterLab offers a more advanced environment with additional features.

INSTALLATION VIA ANACONDA (RECOMMENDED)

[Anaconda](#) is the preferred installation method for most users, as it includes Jupyter and key scientific computing libraries.

INSTALLATION VIA PIP

For those preferring a minimal installation, Jupyter can also be installed using **pip**, the Python package manager.

For detailed instructions, visit the [Jupyter installation guide](#).

Google Colab: A Cloud-Based Alternative

[Google Colab](#) is an alternative to Jupyter that runs entirely in the cloud requiring no installation. It provides free access to computational resources and integrates with Google Drive for storing and sharing notebooks. Google Colab is particularly useful for classroom settings where students may not have access to high-performance local machines.

Key benefits of Google Colab include

- No setup or installation required.
- Free (but limited) access to GPUs.
- Seamless integration with cloud storage for easy data access.
- Built-in support for popular bioinformatics libraries.

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7. Teaching Resources

7.1 Faculty Mentoring Networks: A Community of Practice for Educators

7.1.1 What is a faculty mentoring network?

A Faculty Mentoring Network (FMN) is a structured, collaborative group of educators working together to develop and implement innovative teaching strategies. FMNs provide a supportive environment for faculty to share experiences, discuss challenges, and refine teaching approaches. In the context of Nanopore sequencing, an FMN can help educators navigate the integration of real-world genomics research into their classrooms.

FMNs are an example of a **Community of Practice (CoP)**—a group of people who share a common concern or passion and learn how to improve their practice through regular interaction. CoPs are particularly valuable in education, where faculty may need to acquire new technical skills such as those required for bioinformatics and hands-on sequencing experiments.

7.1.2 Why Join a community of practice?

Educators new to Nanopore sequencing and/or bioinformatics may feel overwhelmed by the breadth of technical knowledge required. A **Community of Practice**, such as a Faculty Mentoring Network, provides:

- **Collaborative Learning** – Educators can share lesson plans, sequencing protocols, and troubleshooting strategies.
- **Expert Guidance** – Experienced mentors provide step-by-step support, making it easier to overcome technical barriers.
- **Peer Support** – Networking with colleagues facing similar challenges fosters a sense of community and shared growth.
- **Classroom Implementation Strategies** – Participants can discuss best practices for integrating sequencing and genomics into undergraduate education**.
- **Resource Sharing** – Members often gain access to teaching materials, datasets, and computational tools developed by the group.



Additional Reading

Adams, S., Tesene, M., Gay, K., Brokos, M., McGuire, A., Rettler-Pagel, T., & Swindell, A. (2023). Communities of Practice in Higher Education: A Playbook for Centering Equity, Digital Learning, and Continuous Improvement. Every Learner Everywhere. <https://www.everylearnereverywhere.org/resources/communities-of-practice-in-higher-education/>.

7.1.3 The Nanopore Faculty Mentoring Network

The **Nanopore FMN**, hosted on **QUBESHub**, is designed to support educators who want to bring Nanopore sequencing into their classrooms**. Participants in this FMN will:

- Gain hands-on experience with **MinION** sequencing and data analysis.
- Work with mentors and peers to design classroom-friendly experiments
- Develop teaching modules that can be adapted to different learning environments.
- Explore strategies for data analysis, storage, and management in an educational setting.

For the latest updates on upcoming mentoring opportunities visit the ****Nanopore Faculty Mentoring Network page**. We also suggest joining the Slack discussion using this **invitation link**.

Joining a FMN is an excellent way to enhance professional development, collaborate with peers, and build confidence in bringing cutting-edge genomics into undergraduate and high school classrooms.

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7.2 Course-Based Research Experiences (CUREs): Transforming STEM Education

7.2.1 What Are Course-Based Research Experiences (CUREs)?

Course-Based Undergraduate Research Experiences (CUREs) integrate authentic scientific research into undergraduate and high school courses, allowing students to engage in real-world inquiry as part of their coursework. Unlike traditional laboratory exercises, which often follow a predetermined set of instructions to verify known scientific concepts, CUREs immerse students in open-ended investigations enabling them to contribute to the broader scientific community.

According to [SERC at Carleton College](#), CUREs have five defining characteristics:

1. **Use of Scientific Practices** – Students formulate hypotheses, design experiments, collect and analyze data, and interpret results.
 2. **Discovery of New Knowledge** – Unlike conventional labs, which confirm established facts, CUREs contribute novel insights to a research field.
 3. **Broadly Relevant Work** – Research projects address unanswered questions that have significance beyond the classroom, often contributing to published research or larger scientific initiatives.
 4. **Collaboration** – Students work in teams to share ideas, troubleshoot problems, and collectively advance the research.
 5. **Iteration** – Students revise their approaches based on their findings, developing resilience and problem-solving skills as they refine their research methods.
-

7.2.2 Why Are CUREs Important?

Integrating research into courses enhances STEM education in several key ways:

- **Increases Student Engagement** – Conducting original research is more stimulating than routine lab exercises, leading to higher student motivation.
 - **Improves Retention in STEM** – Studies indicate that students involved in research are more likely to persist in STEM majors and careers.
 - **Fosters Critical Thinking and Scientific Literacy** – Engaging in research helps students develop skills in data analysis, communication, and problem-solving.
 - **Expands Access to Research** – CUREs enable more students, especially those at minority-serving institutions (MSIs) or under-resourced schools to participate in research experiences that they might not otherwise access.
-

7.2.3 Nanopore Sequencing and CUREs

Nanopore sequencing is well-suited for CUREs in biology and genomics because it allows students to generate and analyze real DNA sequence data within a classroom setting. By incorporating DNA barcoding, metagenomics, or genome sequencing, educators can design research projects that help students:

- **Address authentic scientific questions**, such as identifying environmental microbes or sequencing unknown species.
- **Use cutting-edge sequencing technologies**, mirroring real-world research settings.
- **Develop bioinformatics and data analysis skills**, preparing them for careers in STEM fields.

CUREs that incorporate Nanopore sequencing provide high-impact, interdisciplinary learning experiences exposing students to molecular biology, computational biology, and scientific inquiry.

**Additional Reading**

1. For more information on CUREs and their impact, visit the [CUREnet website](https://serc.carleton.edu/curenet/whatis.html).
2. A Guide to Course-based Undergraduate Research: Developing and Implementing CUREs in the Natural Sciences. First Edition Edition. ISBN-13: 978-1319367183. [Publisher website](#).
3. AAAS (American Association for the Advancement of Science) (ed. C Brewer & D. Smith). (2011). Vision and Change in Undergraduate Biology Education: A Call to Action. ISBN#: 978-0-87168-741-8. <https://visionandchange.org/finalreport/>.
4. Auchincloss LC, Laursen SL, Branchaw JL, Eagan K, Graham M, Hanauer DI, Lawrie G McLinn CM, Pelaez N, Rowland S, Towns M, Trautmann NM, Varma-Nelson P, Weston TJ, Dolan EL. (2017). Assessment of course-based undergraduate research experiences: a meeting report. CBE Life Sci Educ. 13(1):29–40 doi: [10.1187/cbe.14-01-0004](https://doi.org/10.1187/cbe.14-01-0004).
5. Dolan EL. (2016) Course-based Undergraduate Research Experiences: Current Knowledge and Future Directions. Paper commissioned for the Committee on Strengthening Research Experiences for Undergraduate STEM Students, Board on Science Education, Division of Behavioral and Social Sciences and Education, Board on Life Sciences, Division of Earth and Life Studies. Washington, D.C.: National Academies of Sciences, Engineering and Medicine. Available https://sites.nationalacademies.org/cs/groups/dbassesite/documents/webpage/dbasse_177288.pdf.

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7.3 Teaching Resources

This page provides educators with a collection of instructional materials designed to support the integration of Nanopore sequencing into classroom and laboratory settings. Whether you are introducing DNA sequencing concepts, guiding students through hands-on sequencing experiments, or developing bioinformatics skills, these resources will help facilitate engaging and effective teaching.

How to Contribute

We welcome contributions from educators! If you have developed lesson plans, presentations, assessments, or other teaching materials, please consider sharing. To contribute, contact williams@cshl.edu.

7.3.1 PowerPoint Presentations

This section includes downloadable slides that educators can use for lectures and classroom discussions.

7.3.2 Videos

Here, you will find educational videos that explain key concepts and demonstrate laboratory techniques.

7.3.3 Assessments

To gauge student understanding, we provide quizzes, problem sets, and concept questions.

7.3.4 Curricular Plans and Materials

For educators designing course modules or research-based experiences.

7.3.5 Key Papers

This section includes foundational and recent research papers on Nanopore sequencing and its applications in education and research. These papers provide background on the technology, its capabilities, and best practices for implementation in the classroom.

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8. Other Resources

8.1 Frequently Asked Questions (FAQ) for New Nanopore Users

This FAQ addresses common questions educators and first-time users may have about **Nanopore sequencing**, its applications, and how to integrate it into the classroom.

8.1.1 Getting Started

Q: What is Nanopore sequencing?

A: Nanopore sequencing is a method of DNA (RNA and perhaps one day protein) sequencing that detects changes in electrical current as nucleic acids pass through a biological nanopore. This allows for real-time sequencing of long DNA fragments.

Q: What equipment do I need to get started with Nanopore sequencing?

A: The **MinION** device is the most accessible option, along with a computer that meets system requirements, a [flow cell](#), sequencing kits, and basic molecular biology equipment.

Q: How much does it cost to start sequencing?

A: Costs vary depending on the setup. A MinION starter pack (device + [flow cells](#) + sequencing kit) is around \$2,000, but ongoing costs include consumables like [flow cells](#) and reagents. See our [time and costs guide](#).

Q: Do I need prior experience in sequencing or bioinformatics to use Nanopore sequencing?

A: No, but familiarity with basic molecular biology techniques and an understanding of sequencing principles will help. The platform provides user-friendly software such as [MinKNOW](#) and [EPI2ME](#). See also [software guide](#).

Q: How does Nanopore sequencing compare to other sequencing methods?

A: Unlike Illumina sequencing, which provides short, highly accurate reads, Nanopore sequencing produces [long reads](#) that can span structural variants and repetitive regions, with real-time data generation.

8.1.2 Using Nanopore in the Classroom

Q: How can I integrate Nanopore sequencing into my curriculum?

A: Start with simple experiments, such as microbial identification via [16S sequencing](#), and progressively introduce DNA barcoding or small genome sequencing.

Q: Is it safe for students to handle Nanopore sequencing experiments?

A: Yes, with appropriate safety precautions. Many classroom protocols use non-hazardous DNA extraction methods and minimize toxic reagents.

Q: What type of experiments can students do with Nanopore sequencing?

A: Common classroom experiments include: - **Microbiome studies** (16S sequencing) - **DNA barcoding** (identifying species using specific gene markers) - **Small genome sequencing** (e.g., bacteriophage or bacterial genomes)

Q: How long does a typical sequencing experiment take?

A: From sample preparation to sequencing: - **DNA extraction:** 1-2 hours - **Library preparation:** 1-2 hours - **Sequencing:** 1-24 hours (depending on read depth) - **Data analysis:** 1-3 hours These are general answers, check your specific protocol for more details.

Q: Do students need their own sequencing devices?

A: No, a single [MinION](#) device can be shared, with students processing different samples in barcoded multiplexed runs.

8.1.3 Sequencing and Data Handling

Q: What kind of samples can be sequenced?

A: Almost any biological sample containing DNA, including: - Environmental samples (soil, water, surfaces) - Microbial cultures - Human samples (with appropriate ethics approval) - Plant or animal tissues

Q: How much DNA is required for sequencing?

A: This depends on the kit:

- **Rapid barcoding kit:** ~10 ng per sample
- **Ligation sequencing kit:** ~100 ng per sample
- **16S sequencing kit:** ~10 ng per sample

Q: How is sequencing data stored and analyzed?

A: Sequencing generates [PODS](#) files (raw signal data) and [FASTQ](#) files (nucleotide sequences). Data can be analyzed using [EPI2ME](#), Galaxy. See also [data guide](#).

Q: What storage is needed for sequencing data?

A: A minimum of 500GB storage is recommended, with 1TB preferred. Cloud-based solutions like CyVerse, JetStream2, or AWS S3 are also options. See also [data guide](#).

Q: Can Nanopore sequencing be used for metagenomics?

A: Yes, Nanopore sequencing can be used for metagenomic studies, enabling long-read assembly of complex microbial communities.

8.1.4 Troubleshooting and Maintenance

Q: What if my sequencing run fails?

A: Common issues include:

- **Low DNA input** → Ensure enough high-quality DNA is used.
- **Flow cell clogging** → Use a **flush kit** to unclog pores.
- **Low sequencing yield** → Check sample preparation and quality.

Q: How should flow cells be stored?

A: Flow cells should be **kept at 2-8°C** and used before the **expiration date**. If not used immediately, they should not be frozen.

Q: How do I clean and reuse a flow cell?

A: [MinION Flow cells](#) can be washed and reused 1-2 times. For another sequencing run you must use a Flow Cell Wash Kit. [Flongle flow cells](#) are one-time use.

Q: How long does a flow cell last?

A: A [MinION flow cell](#) typically lasts for 1-2 sequencing runs with an 11-week shelf life. [Flongles](#) typically last 4-6 weeks in our experience.

8.1.5 Advanced Topics

Q: Can I run multiple samples at once?

A: Yes, [multiplexing](#) allows you to sequence multiple barcoded samples in the same run, reducing costs.

Q: Can I sequence RNA instead of DNA?

A: Yes, **direct RNA sequencing** is possible with Nanopore, allowing you to study full-length transcripts and modifications. As of yet, we don't cover this topic as the specialized requirements for working with RNA is likely less common in general classroom settings.

Q: Can I use cloud computing for sequencing analysis?

A: Yes, [JetStream2](#), [AWS](#), and [EPI2ME](#) offer cloud-based solutions for [basecalling](#) and data analysis.

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8.2 Nanopore Sequencing Glossary

This glossary provides definitions of key terms related to **Nanopore sequencing**, offering educators and first-time users a foundational understanding of the technology and its associated concepts.

8.2.1 A

Adapter

Short DNA or RNA sequences ligated to sample fragments to facilitate their attachment to sequencing platforms.

Alignment

The process of arranging sequences to identify regions of similarity, which may indicate functional, structural, or evolutionary relationships.

Amplicon

A DNA fragment that has been amplified by PCR.

8.2.2 B

Base Modification

Chemical alterations to DNA bases, such as methylation, which can affect gene expression.

Basecalling

The process of determining the sequence of nucleotides (bases) from raw signal data generated by the sequencing platform.

Barcode

Short, unique sequences added to DNA fragments to identify and differentiate multiple samples sequenced together. See also **Indexing**.

8.2.3 C

Chimera

A sequence artifact formed by the joining of two or more unrelated DNA fragments.

Consensus Sequence

A sequence derived from the alignment of multiple sequences, representing the most common nucleotide at each position.

Contig

A contiguous sequence assembled from overlapping reads.

Coverage

The number of times a nucleotide is read during sequencing; higher coverage increases confidence in the accuracy of the sequence.

8.2.4 D

De Novo Sequencing

Sequencing a novel genome without a reference, assembling the sequence from scratch.

Demultiplexing

The process of sorting sequences from a mixed sample pool into individual datasets based on their barcodes.

Direct RNA Sequencing

Sequencing RNA molecules directly without converting them to cDNA, preserving base modifications.

DNA Library

A collection of DNA fragments prepared for sequencing.

Dorado

A basecalling software developed by Oxford Nanopore Technologies for translating raw nanopore signals into nucleotide sequences.

8.2.5 E

Epigenetics

The study of heritable changes in gene expression that do not involve changes to the underlying DNA sequence.

Epigenome

The complete set of epigenetic modifications on the genetic material of a cell.

8.2.6 F

FAST5

A file format previously used by ONT to store raw signal data from nanopore sequencing. It has been replaced by **POD5**.

FASTQ

A common file format for storing nucleotide sequences and associated quality scores.

Flow Cell

A consumable device containing nanopores through which nucleic acids pass during sequencing.

8.2.7 G

8.2.8 H

High Molecular Weight (HMW) DNA

Long DNA fragments essential for obtaining long reads in sequencing.

8.2.9 I

Indexing

The use of unique sequences (indexes), also called barcodes, to tag DNA fragments from different samples, enabling their identification after pooled sequencing.

8.2.10 L

Library Preparation

The process of preparing DNA or RNA samples for sequencing, including fragmentation, adapter ligation, and amplification.

Ligation

The enzymatic process of joining two DNA fragments together.

Long Reads

Sequences that are typically longer than 10,000 base pairs, advantageous for resolving repetitive regions and structural variants.

8.2.11 M

MinION

A portable nanopore sequencing device developed by Oxford Nanopore Technologies.

Multiplexing

Sequencing multiple samples simultaneously by using unique barcodes for each sample.

8.2.12 N

Nanopore

A nanometer-scale pore used in sequencing to detect changes in ionic current as nucleic acids pass through, allowing base identification.

Oxford Nanopore Technologies (ONT)

The company that developed and commercialized nanopore sequencing technology.

8.2.13 P

Pore

A tiny opening in a membrane through which molecules can pass.

PODS

The current file format used by ONT to store raw sequencing signal data.

PromethION

A high-throughput nanopore sequencing device designed for large-scale projects.

8.2.14 Q

Quality Score (Phred Score)

A metric that indicates the confidence of a base call; higher scores represent greater confidence.

8.2.15 R

Rapid Adapter

Short DNA sequence ligated to sample fragments to facilitate their translocation through the nanopore.

Read

A sequence of nucleotides generated by a sequencing machine from a single DNA or RNA fragment.

Real-Time Sequencing

The ability to analyze sequencing data as it is generated, enabling immediate insights.

Reference Genome

A representative example of a species' genome used as a standard for comparison.

8.2.16 S

Single Molecule Sequencing

Sequencing technologies that read individual DNA or RNA molecules without amplification.

8.2.17 T

Throughput

The amount of data generated by a sequencing platform in a given time frame.

Translocation

The movement of nucleic acid molecules through a nanopore during sequencing.

8.2.18 U

Ultra-Long Reads

Sequences exceeding 100,000 base pairs, useful for assembling complex genomes.

8.2.19 V

Variant Calling

The process of identifying variations (e.g., SNPs, insertions, deletions) between sequences and a reference genome.

8.2.20 Y

Yield

The total amount of data (in bases) produced by a sequencing run.

8.2.21 Z

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9. Contribute

9.1 Contributions

This e-book is intended to be a collaborative resource developed by and for the community of educators using Nanopore sequencing in their classrooms and laboratories. As more instructors integrate real-world DNA sequencing experiences into their courses, sharing materials, insights, and best practices helps build a stronger foundation for effective, accessible, and innovative teaching.

We welcome contributions from educators, researchers, and practitioners to expand and refine this resource. There are many ways to contribute, ranging from submitting instructional materials to providing feedback on existing content.

9.1.1 Ways to Contribute

Provide feedback

If you've used materials from this e-book and found areas for improvement, or if you'd like to suggest new topics, your feedback is valuable. Educators who are actively using Nanopore sequencing with students can help ensure that our content remains:

- **Practical** – Aligned with real-world classroom and laboratory settings
- **Accurate** – Updated with the latest scientific knowledge and best practices
- **Inclusive** – Designed to be accessible to a diverse range of students

To share feedback options include:

1. Email williams@cshl.edu
2. Participate in the discussion at [QUBES Nanopore Network](#) or [Nanopore Network Slack](#).
3. Visit the website's [GitHub page](#) and submit an [issue](#).

Share teaching resources

If you have developed lesson plans, presentations, worksheets, videos, or assessments, we encourage you to share them with the community. Contributions may include:

- **Lecture slides or presentations**
- **Step-by-step protocols** for DNA extraction, sequencing, or data analysis
- **Sample datasets for classroom use**
- **Assessments or concept questions** to test student understanding
- **Complete course modules or research experiences (CUREs)**

Submit research and case studies

If you have conducted research on Nanopore sequencing in the classroom or developed a case study about how it has impacted student learning, we encourage you to share your findings. Potential contributions include:

- Published papers or preprints on Nanopore-based education initiatives
- Case studies of student-led sequencing projects
- Institutional reports on course-based research experiences (CUREs)

Contribute to the glossary or FAQs

The [glossary](#) and [FAQ](#) sections are designed to help newcomers quickly understand technical terms and common questions about Nanopore sequencing. If you notice missing terms or important questions that should be addressed, your contributions can enhance clarity for future educators and students.

9.1.2 Contribution templates

You can visit examples/protocol-template.md to see and download a template for contributing protocols.

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